

Technology Complementarities and Subsidy Policy: Evidence from Electric Vehicle and Solar Panel Adoption

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Abstract

Government policies target air pollution and climate change by incentivizing adoption of electric vehicles (EVs) and/or residential solar panels (PVs). Knowledge of whether these goods are complements or substitutes can be used to design policies that target environmental externalities more efficiently. I use California household-level data to estimate a structural multi-product demand model. I find that consumers view PVs and EVs as complements, with the degree of complementarity varying with vehicle size and income. Counterfactual experiments reveal that complementarity significantly increases bundled EV-PV purchases. This complementarity can be leveraged to design policies that achieve emission targets at lower cost.

JEL: D12, Q42, Q48, Q51, Q54, Q58, R48.

Keywords: Solar Panels, Electric Vehicles, Complementarity, Multi-Product Demand Model, Environmental Policy, Renewable Energy.

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I Introduction

Transportation and electricity generation are two of the main sources of greenhouse gas emissions and local air pollution (U.S. Environmental Protection Agency, 2022). Governments around the world devote substantial resources to decarbonizing these sectors. In the United States, for example, federal tax expenditures on electric vehicles and residential solar panels exceeded \$20 billion in 2024 alone (U.S. Department of the Treasury, 2024).

Most economic analyses of these technologies treat the markets for electric vehicles (EVs) and solar photovoltaic (PV) systems as independent. However, the two technologies may interact to jointly affect consumer demand. Identifying how consumers perceive and value the relationship between EVs and PVs is important for designing subsidies that achieve a targeted reduction in emissions at the lowest possible cost.

In this paper, I develop a multi-product discrete-choice demand model with flexible substitution patterns to study the interaction between EVs and PVs. This framework enables me to assess the efficiency of various counterfactual policies aimed at promoting the adoption of EVs and PVs in terms of their ability to mitigate environmental externalities, offering new insights into the design of coordinated clean-technology subsidies.

Electric vehicles and solar panels may be complements or substitutes. The potential channels for complementarity include lower operating costs, driven by specific electricity rate structures, and environmental preferences, as the environmental benefits of driving an EV may be perceived to be amplified when the household owns rooftop solar, since solar generation offsets part of grid consumption, including that associated with EV charging, whether or not the vehicle draws on solar generation at the moment it is charged.

The products could, however, be substitutes. A potential mechanism for substitutability is that both goods satisfy the demand for environmentally friendly technology or the desire to signal a “green” lifestyle. Moreover, both technologies enable long-term cost savings and can satisfy preferences for new technology. As a result, budget-constrained consumers may select only one technology to suit their preferences.

I focus on California, which is the state with the most solar energy production, EV sales, and public EV charging stations (Environment California Research and Policy Center, 2021). My primary data set is the California Vehicle Surveys in 2013, 2017, and 2019. Respondents report their vehicles’ features, including size, fuel type, purchase year, car model year, and whether they have solar panels or plan to install them in the next five years. I also observe each respondent’s age, income, type of housing, household size, and education.

I use these data to estimate a static discrete-choice model of demand for solar panels and electric vehicles following Gentzkow (2007). The model allows consumers to purchase a car only,

a solar panel system only, or both, without imposing any a priori assumption about whether EVs and PVs are substitutes or complements. There are two key empirical challenges in identifying the model. The first challenge is the endogeneity of prices, which arises because the unobserved characteristics of the product that influence demand are also correlated with prices. To address this issue, I use the instrumental variable approach. Specifically, I instrument for prices using the sums of characteristics of competing products in the same market. The instrument's design follows the logic of Berry, Levinsohn and Pakes (1995), which holds that a product's price will depend on its comparability to other goods in the market. The second challenge is disentangling the degree of complementarity between EVs and PVs from correlated consumer preferences for these technologies. I address this by controlling for observable consumer demographics, which account for systematic differences in preferences (e.g., income, education, household size, or age), and by using product-specific demand shifters that satisfy exclusion restrictions, such as the number of public charging stations and solar irradiation, which shift the utility of one product without directly affecting the other. I use a two-stage procedure to estimate the structural parameters following Berry, Levinsohn and Pakes (1995) and Bayer, Ferreira and McMillan (2007).

I find that the average household values higher gas mileage, greater horsepower, and the adoption of solar systems, while experiencing disutility from higher purchase prices and increased operational driving costs. Additionally, the results suggest that solar panels and electric vehicles are complements, with the willingness to pay for this complementarity increasing with income. On average, a 1% decrease in the price of EVs results in a 0.23% increase in the demand for PV systems. On the other hand, a 1% decrease in the price of PV systems results in a 0.69% increase in the demand for EVs.

I use these results to conduct three counterfactual experiments. First, I quantify the contribution of complementarity between PVs and EVs to the adoption of each technology at current prices. I find that complementarity plays a substantial role in the joint take-up rate. In the absence of complementarity, the purchases of EVs and PVs together would be 64% lower.

Second, I consider the consequences of ignoring complementarity when designing subsidies aimed at reducing emissions. I show that subsidies for solar panels are more effective than those for EVs in reducing fine particulate matter ($PM_{2.5}$), while EV subsidies reduce more carbon dioxide (CO_2): a \$2 billion subsidy to one technology alone reduces CO_2 by 1.49 million metric tons for EVs against 1.00 for PVs, and $PM_{2.5}$ by 47 metric tons for PVs against 20 for EVs. However, complementarity further improves the effectiveness of both PV and EV subsidies, with the amplification particularly strong for EV subsidies: 29% of the CO_2 reductions and 43% of the $PM_{2.5}$ reductions attributed to EV subsidies arise from the complementary adoption of solar panels.

Finally, in the third experiment, I investigate the efficient way to allocate a fixed budget for

subsidies between the two markets. I find that the optimal subsidy allocation depends on the relative weighting of local and global pollutants. Allocating approximately 90% of the budget to PV subsidies maximizes $PM_{2.5}$ emissions reductions, while allocating 27% to PV minimizes CO_2 emissions most efficiently. When weighting the two pollutants using U.S. government valuations of their total social benefits, the optimal allocation is 29% to PV subsidies and 71% to EV subsidies.

Overall, the paper contributes to the growing literature on PV and EV adoption. Prior studies investigated how adoption of EVs is affected by policy incentives (Barwick, Kwon and Li, 2024; Muehlegger and Rapson, 2023, 2022; Armitage and Pinter, 2022; Remmy, 2023; Xing, Leard and Li, 2021; Jenn, Springel and Gopal, 2018; Sheldon, DeShazo and Carson, 2017), deployment of the public charging infrastructure (Dorsey, Langer and McRae, 2022; Springel, 2021; Li, 2023; Li et al., 2017), demographic heterogeneity and political ideology (Davis, Li and Springel, 2023; Jacqz and Johnston, 2023; Archsmith, Muehlegger and Rapson, 2021; Linn, 2022; Sheldon and Dua, 2019), peer effects (Tebbe, 2023), and technology advancement (Forsythe et al., 2023). Previous studies of rooftop PV adoption examined the effects of financial incentives (Feger, Pavanini and Radulescu, 2022; Langer and Lemoine, 2022; Pless and van Benthem, 2019; De Groote and Verboven, 2019; Gillingham and Tsvetanov, 2019; Burr, 2016; Hughes and Podolefsky, 2015; Sarzynski, Larrieu and Shrimali, 2012), heterogeneity of adopters' socioeconomic characteristics (Dorsey and Wolfson, 2024; De Groote and Verboven, 2019; Crago and Chernyakhovskiy, 2017; De Groote, Pepermans and Verboven, 2016; Borenstein and Davis, 2016; Kwan, 2012), pro-environmental preferences (Crago and Chernyakhovskiy, 2017; Dastrup et al., 2012), and peer effects (Bollinger et al., 2022; Gillingham and Bollinger, 2021). Furthermore, Bollinger et al. (2023) consider complementarity between rooftop solar and energy storage technology adoption. Additionally, several studies investigate spatial variation in the environmental benefits of EV and PV adoption (Dauwalter and Harris, 2023; Graff Zivin, Kotchen and Mansur, 2014; Muehlegger and Rapson, 2020; Holland et al., 2016).

Although several studies examine PV and EV adoption independently, the relationship between the two technologies has received relatively little attention. Prior studies provide evidence of a positive relationship between EV and PV adoption (Delmas, Kahn and Locke, 2017; Cohen et al., 2019; Ferdousee, 2021; Lyu, 2023; Sharda et al., 2024). In particular, Lyu (2023) provides causal evidence that adoption of each technology increases adoption of the other. I build on this evidence by estimating a structural multi-product demand model that quantifies complementarity directly in household utility and characterizes how it varies across consumers and vehicle types. This paper therefore contributes to the literature by linking the estimation of utility complementarity to the design of subsidy policy across interrelated clean technologies. Recovering the underlying demand system allows me to quantify cross-price responses and evaluate counterfactual subsidy policies that account for interactions between the EV and PV markets and their common externalities. I show

that accounting for these cross-market interactions can significantly improve the cost-effectiveness of clean-energy programs.

The next section provides background on California’s solar and electric vehicle industries. Section III describes the data and presents reduced-form estimates of the relationship between solar panel systems and electric car adoption. Section IV specifies the demand model with complementarity for two technologies followed by identification arguments; Section V outlines the estimation strategy. The results are described in Section VI. Section VII presents counterfactual policy evaluations. Section VIII concludes.

II Background

A Electric Vehicle Industry

Electric vehicles include battery-electric vehicles (BEVs), which only run on electricity; plug-in hybrid electric vehicles (PHEVs), which can use both an electric motor and an internal combustion engine as a backup; fuel cell electric vehicles (FCEVs), which use an electric-only motor powered by hydrogen; and plug-in fuel cell electric vehicles (PFCEVs), which combine features of BEVs and FCEVs. According to the California Energy Commission (2024), California sold 1,996,931 total EVs as of the second quarter of 2024, the most of the 50 states.¹ It accounted for roughly 39% of all new electric car sales between 2011 and 2024 in the United States.

Electric vehicle incentives in the United States have been in place for over ten years. The incentives come in the form of rebates on purchases, tax exemptions and credits, and other benefits, such as access to HOV lanes and fee waivers (charging, parking, tolls, etc.).² The size of a car’s battery or the range of an all-electric vehicle may affect the amount of financial incentives.

A federal income tax credit of up to \$7,500 is available for BEV and PHEV vehicles acquired in 2010 or later.³ I take the federal subsidy into account when constructing the prices of electric vehicles in my analysis.

Moreover, there are various incentives provided by states and utility companies.⁴ However, I do not account for these rebates in my analysis because I am unable to observe take-up of the rebates

¹Figure A.1 shows the electric vehicle market has been steadily growing since its inception in the late 2000s and is projected by industry analysts to expand more in the following years.

²California’s Clean Air Vehicle (CAV) decal program, which allowed solo drivers of qualifying electric vehicles to use HOV lanes, expired on September 30, 2025; CAV decals could no longer be used to access HOV lanes as of October 1, 2025 (California Department of Motor Vehicles, 2025).

³The federal electric vehicle tax credits were eliminated by the One, Big, Beautiful Bill Act, signed into law on July 4, 2025. The act terminated the credits for new, used, and commercial electric vehicles purchased after September 30, 2025.

⁴For example, California’s Clean Vehicle Rebate Project (CVRP), created in 2010, offered up to \$2,500 for the purchase or lease of BEVs and \$1,500 for the purchase or lease of PHEVs. Low-income households that meet the requirements could qualify for an extra subsidy of \$2000 starting from 2016.

and whether individual households satisfy eligibility criteria which are linked to vehicle models, household income, and varied features of regional utility programs. This omission introduces a likely source of measurement error in vehicle prices, which I address through an instrumental variables approach.

B Solar Photovoltaic (PV) Industry

California is an attractive place for property owners to install solar panels because of its pro-solar laws, abundant sunshine, and generous solar subsidies. California consistently ranks first in the nation for producing solar energy. It has the highest installed capacity out of all 50 states and ranks second in solar installations per capita, surpassed only by Hawaii. 25% of the state's electricity is generated from solar.⁵

Several incentive programs were in effect during my study period. One of the most important federal policies promoting the expansion of solar energy in the US is the 30% solar Investment Tax Credit (ITC) implemented in 2006 (Solar Energy Industries Association, 2022).⁶

In addition to the federal ITC, California offers other solar incentive programs, depending on the location and utility provider. For example, the CSI Single-Family Affordable Solar Homes (SASH) Program has provided a capacity-based incentive of \$3,000 for every kW of home solar installed to qualified low-income households since 2009 (California Public Utilities Commission, 2022a).

Another important policy is net energy metering (NEM). During the analyzed period, NEM 1.0 and NEM 2.0 were in effect. NEM allows customers of three utilities (PG&E, SCE, and SDG&E) to produce their own energy to serve their electricity needs directly on-site and obtain a credit toward their electric bills for any excess energy sent back to the grid. Customers that generated electricity for the grid received full retail rate credits (California Public Utilities Commission, 2022b).

C Substitutes or Complements?

In principle, solar panels and electric vehicles could be substitutes or complements. One potential mechanism for complementarity is decreasing operating costs when used together. Many utilities provide electricity price plans that make coordinated use of solar generation and EV charging more cost-effective. Appendix F provides examples.

Another potential channel is the preference for being environmentally friendly. In this case, PV adoption increases the utility from owning an EV because owners derive satisfaction from knowing

⁵The left panel of Figure A.2 shows California's mean installed price per kW from 2007 to 2020. Between 2007 and 2020, prices fell by over 40%. The right panel demonstrates the total solar capacity installed each year between 2007 and 2019.

⁶On July 4, 2025, the One, Big, Beautiful Bill Act was signed into law, eliminating the federal solar tax credit for homeowners effective December 31, 2025.

that the environmental benefits of EV ownership depend on the cleanliness of electricity generation (Holland et al., 2016). When a household owns solar, the perceived environmental benefit of driving an EV rises, reinforcing complementarity between the two technologies. Whether the physical benefit rises as well depends on when charging occurs relative to generation: a vehicle charged overnight draws on the same marginal generator whether or not the household owns a panel, while the panel's output displaces grid generation either way. The emission calculations in Appendix H therefore do not credit joint ownership with any reduction beyond what each technology achieves separately. What the complementarity parameter captures is the value households place on joint ownership, which may rest on the perceived benefit whether or not the physical one is present.⁷

On the other hand, there are at least three channels through which EVs and PVs could be perceived as substitutes. First, both goods may satisfy the demand for “green” products. Due to the high costs of adopting these technologies, budget-constrained households may choose to adopt just one of the two technologies to reduce their carbon footprint (Kahn, 2007). Similarly, both goods can serve to signal social status, the owner's “green” lifestyle, or an orientation toward innovation (Griskevicius, Tybur and Van den Bergh, 2010; Sexton and Sexton, 2014). Finally, both technologies enable long-term cost savings on energy or gasoline bills despite high upfront costs. Hidrue et al. (2011) find that savings are the primary motivation for most people to buy an EV. Again, budget-constrained households may choose to invest in only one technology.

The literature provides some evidence that EVs and PVs may act as complementary technologies. Delmas, Kahn and Locke (2017) provide descriptive evidence of increasing joint purchases of EVs and PVs in California. Ferdousee (2021) uses a bivariate probit regression model to show that education, income, and household type are associated with the probability of joint adoption in California. Lyu (2023) uses an instrumental variables regression to conclude that each existing PV leads to approximately 0.184 additional EV sales, while each EV purchase leads to about 0.26 additional PV installations. Cohen et al. (2019) find that having an EV in Austria increases the likelihood of owning a PV by 31 percentage points, and the probability of owning an EV increases by 7.1 percentage points for PV owners. While these studies establish a positive relationship between EV and PV adoption, they do not recover the structural demand parameters needed to evaluate counterfactual subsidy policies. This study advances the literature by developing the first discrete-choice model that structurally identifies complementarity (or substitutability) between electric vehicle and

⁷For example, Coffman, Bernstein and Wee (2017) find that given Hawaii's electricity fuel mix, the Nissan Leaf (EV) would produce 31 $MTCO_2$ and the Toyota Prius (hybrid electric vehicle) 27 $MTCO_2$ during their 150,000-mile lifetimes. However, under direct daytime charging from residential solar PV on weekends, the GHG emissions of EVs fall substantially relative to HEVs. In this case, the Nissan Leaf will produce almost 8 $MTCO_2$ less than the Toyota Prius over their lifetimes. This channel requires charging to coincide with generation. The charging profile assumed throughout this paper places EV charging off-peak overnight (Appendix G), so neither the savings nor the emission calculations rely on it.

solar panel purchases. The framework allows for flexible substitution patterns between the two sides of the market, quantifies the degree of complementarity, and enables counterfactual policy evaluation.

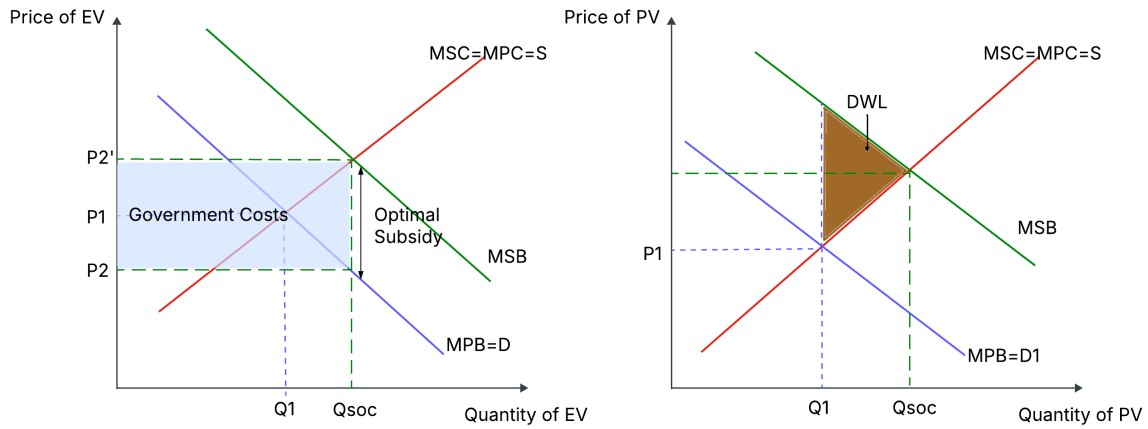
To illustrate the policy relevance of these interactions, consider the stylized supply and demand diagrams presented in Figure 1. They demonstrate how the effect of a subsidy for the positive externalities of EVs on the PV market varies depending on whether the two products are independent, substitutes, or complements.

Both EVs and PVs produce positive externalities by reducing CO_2 and $PM_{2.5}$ emissions. Let Q_{soc} denote the socially optimal quantity for EVs and PVs in their respective markets. Panel (a) illustrates the case where demand for the two goods is independent. Here, a subsidy for EVs decreases the consumer's price from P_1 to P_2 , resulting in an increase in the quantity of EVs demanded from Q_1 to the socially optimal level (Q_{soc}). P_2' here reflects the new total (pre-subsidy) price of the product. Importantly, the demand for PVs remains unaffected as these products are independent. The government cost to achieve this result is indicated by the shaded blue area in the left panel. The brown shaded region in the right panel highlights the deadweight loss (i.e., the welfare loss relative to the social optimum), which differs across the three scenarios depending on whether EVs and PVs are independent, substitutes, or complements.

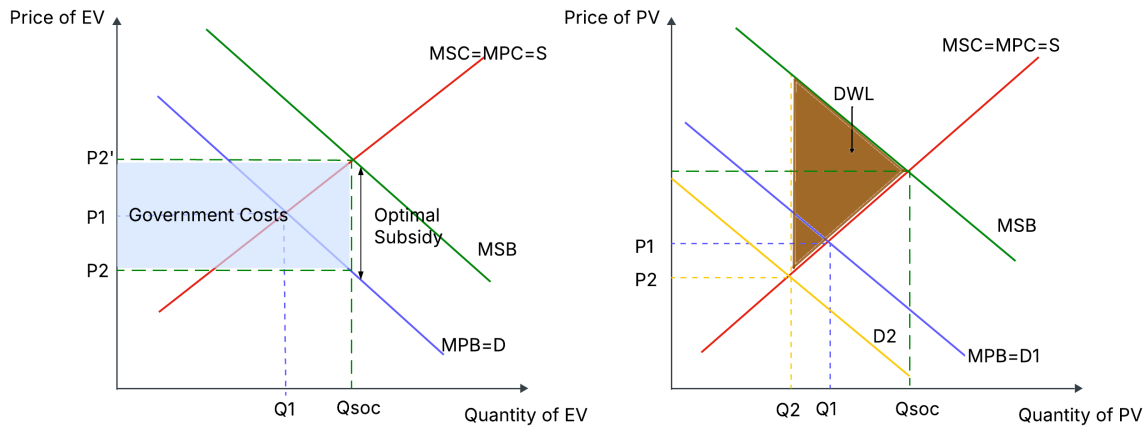
In the case of substitutes, panel (b) shows that as the price of EVs decreases due to a subsidy, the demand for PVs shifts leftward from D_1 to D_2 . This substitution effect diminishes the overall effectiveness of carbon reduction policies by reducing solar installations and, therefore, weakens the intended externality reduction. The deadweight loss expands. Consequently, the government must increase subsidies for both products to maintain the socially optimal adoption levels of each.

Finally, panel (c) illustrates the complementarity scenario, where EV subsidies not only increase EV adoption, but also increase demand for photovoltaics, shown by a rightward shift from D_1 to D_2 in the PV market. Therefore, now the government has to spend fewer resources to achieve a socially optimal level of PV adoption. The complementarity effect reinforces the subsidy's impact, suggesting that well-coordinated policies can exploit this relationship to maximize reductions in emissions.

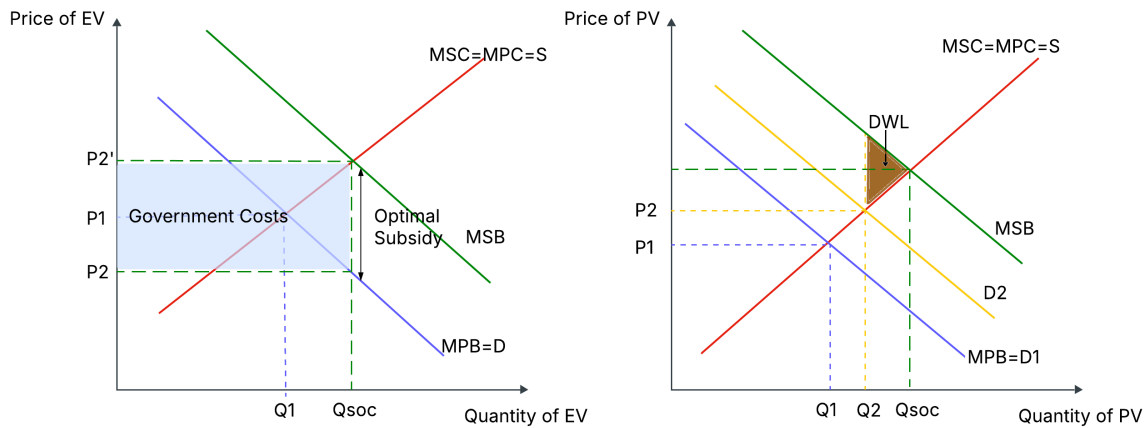
In this paper, I apply a multi-product model to quantify how interactions between EVs and PVs shape the effectiveness of environmental policy. Accounting for these cross-technology interactions is essential, as subsidies in one market can induce adoption in the other. By capturing total rather than market-specific externalities, the model shows that emission targets can be achieved with fewer resources than if the goods were assumed independent.



a) Independent Goods



b) Substitute Goods



c) Complementary Goods

Figure 1: Optimal Subsidies in Interdependent Markets

Note: These figures illustrate how a subsidy in one market (e.g., EVs) affects the other market (e.g., PV) under three scenarios. When goods are independent, the subsidy has no effect on the other market. When goods are substitutes, the subsidy increases deadweight loss in the other market, as higher demand for the subsidized good reduces demand for its substitute. When goods are complements, the subsidy reduces deadweight loss in the other market by amplifying joint demand.

III Data

A Data Sources

My primary data source is the California Vehicle Survey (CVS) of household and commercial vehicle fleet owners conducted by the California Energy Commission in 2013, 2017, and 2019 (National Renewable Energy Laboratory, 2024). The survey is a repeated cross-section: each wave samples a new set of respondents. The data describe individual and household characteristics of respondents and their ownership of the technology. In particular, I observe whether a household has solar panels installed on the residence, whether they are planning on purchasing solar panels within the next five years, and characteristics of the respondent’s car.⁸ I restrict the sample to respondents who reside in single-family or mobile houses because these households can install a PV system on their property. The 2010–2012 California Household Travel Survey (CHTS) was used to get data on the 2013 CVS respondents’ vehicles. Households that completed the CHTS and stated their intention to purchase a vehicle in the near future were included in CVS’s household component. Both surveys used the same household ID numbers, enabling their responses to be linked.

The linked data overrepresent individuals who are highly educated and have higher incomes, while minority groups are underrepresented. Additionally, the survey slightly overrepresents those in the 35–64 age group, who make up 59% of respondents compared to 53% in the broader population. Participants are representative of the geographic distribution of households throughout California, and roughly representative of “1, 2, and 3 or more” household vehicle ownership categories as reported in the 2015 American Community Survey. In Section VII, I apply sampling weights to ensure that the simulation of counterfactual policies is representative of California’s demographics.⁹

The second data source is the Distributed Solar Public Data from Lawrence Berkeley National Laboratory (2023) (LBNL). LBNL publishes non-confidential project-level data on residential photovoltaic systems. The data include the total installed price for the system, the installation date, the system size, zip code, customer segment, and other system features. I used their data to construct prices of solar panels in different geographic areas and years.

To construct vehicle prices, I used several datasets. The vehicle MSRP prices were taken from the AutoWeb (2022) website. California Auto Outlook reports published by California New Car Dealers Association (2022) provide annual top-selling models in each segment and their total registrations in California. I used these reports to construct average prices for gas and hybrid

⁸Respondents report the purchase year of each vehicle but not the year a solar system was installed, so the order in which a household acquired the two technologies is not observed.

⁹Table A.1 presents summary statistics from the combined three California Vehicle Surveys, along with comparative data from the population-representative Current Population Surveys (CPS) conducted in 2013, 2017, and 2019 (Flood et al., 2023). The statistics in both surveys are presented for people who live in single-family or mobile houses.

vehicles in each size segment each year. The California Energy Commission provides statistics on annual new Zero Emission Vehicles sales in California (California Energy Commission, 2022). I used this dataset to construct average electric vehicle prices by different size categories for different years. In addition, I accounted for electric vehicle federal incentives of \$7,500 in the product price.¹⁰

Global horizontal irradiation variables for California counties were retrieved from Global Solar Atlas (Solargis, 2022). The number of charging stations in each county for each of the analyzed years was taken from the Alternative Fuels Data Center by the U.S. Department of Energy (2022). The essential variables that this dataset contains are the addresses of all charging stations for electric vehicles, dates when they were opened, and whether they are publicly accessible. The number of high-occupancy vehicle lanes in different regions of California was drawn from the annual reports of the California Department of Transportation (2022).

B Descriptive Evidence

Figure 2a demonstrates the proportion of individuals who own each technology for each year the survey was implemented. Most respondents (approximately 81%) have neither electric vehicles nor solar panels. However, the percentage of respondents who own one or both technologies has been rising over time.

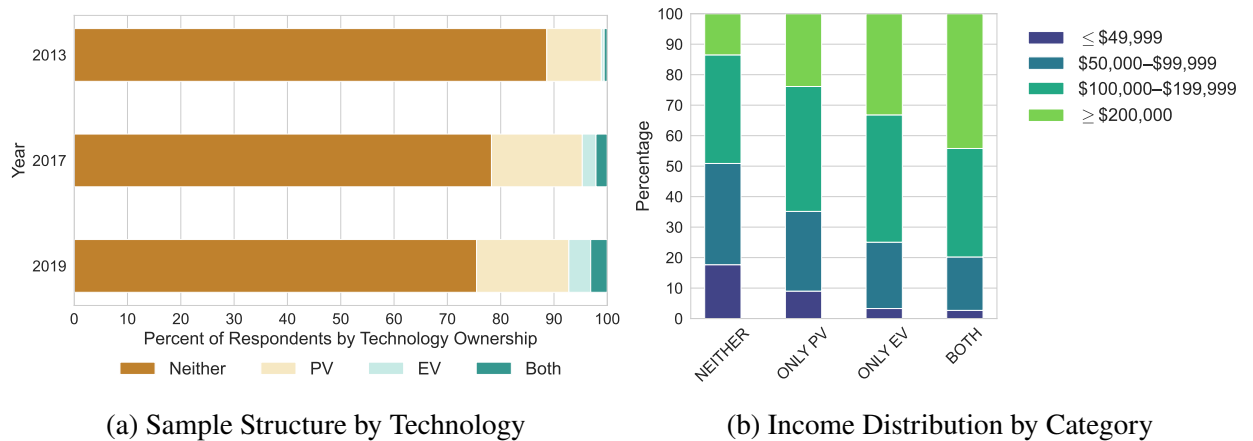


Figure 2: Technology Ownership by Year and Income

The sample contains significantly more PV owners than EV owners. A potential reason for this pattern could be the relative maturity of the two markets. Solar incentives have been in place since 2006, and by the time of the sample, installation costs had already declined substantially and

¹⁰As noted earlier, I did not include state and utility company level incentives due to varying eligibility requirements and my inability to observe take-up. This introduces some measurement error in prices which I address using IVs explained in section V.

adoption was visible across many California neighborhoods. By contrast, EV incentives were only introduced in 2010, and in the early 2010s the EV market was still emerging, with limited model availability, shorter ranges, and sparse charging infrastructure.

Figure 2b shows how household income varies with technology adoption. As expected, mean household income increases as we move from non-ownership to those owning a PV only, to those owning EVs only, and finally to those owning both technologies.

According to the survey, 68% of EV owners have installed PV or intend to install it within the next five years, which is consistent with the hypothesis that these two technologies complement each other.¹¹ In order to quantify the difference in probabilities of purchasing a PV for households with and without electric cars and vice versa, I estimate the following logit regression:

$$P(Y = 1 \mid X_1, X_2, \dots, X_K) = \frac{1}{1 + e^{-(\beta_0 + \sum_{k=1}^K \beta_k X_k)}} \quad (1)$$

where Y equals 1 if the household owns the technology and 0 otherwise, and X_k ($k = 1, \dots, K$) are the independent variables. Explanatory variables include ownership of a PV (or an EV, depending on the regression), as well as controls for age and income groups, household size, and county and year fixed effects.¹² I find that households with a PV system have 3.28 times higher odds of owning an EV compared to households without a PV system, all else constant. Similarly, households that own an EV have 3.13 times higher odds of having a PV system relative to households without an EV. Odds ratios for both EV and PV ownership are monotonically increasing with income. Additionally, larger households are more likely to invest in PV systems but less likely to purchase EVs.¹³

Rescaling these results as average marginal effects reveals that the probability of purchasing an EV is 5.7 percentage points higher if the household owns a PV, while EV ownership is associated with a 19.3 percentage point increase in the probability of installing a PV. These results are similar to the evidence for Austria from Cohen et al. (2019), who report average marginal effects of 7.1 and 31 percentage points. The findings of Lyu (2023) are also qualitatively similar but a direct comparison is complicated by differences in effect size measures and study populations.¹⁴

Thus, adoption of the two technologies is strongly positively correlated. A household that owns one technology is, on average, more likely to own the other. This positive relationship could arise because solar panels and electric vehicles are complements, but it could also reflect correlated

¹¹The survey separately asks whether respondents intend to install PV within the next five years. This descriptive statistic includes both current owners and respondents reporting installation intentions. The descriptive regressions and structural estimation use current ownership only; intended installations are not coded as ownership.

¹²As more control variables are included, the coefficient on technology ownership decreases but continues to indicate a large and precisely estimated positive association.

¹³Full regression results are reported in Tables B.2 and B.3.

¹⁴Average marginal effects and standard errors are reported in Table B.4.

preferences for the two technologies. Lyu (2023) provides complementary causal reduced-form evidence using an instrumental-variables design and finds that adoption of either technology increases adoption of the other. The analysis below builds on this evidence by using household-level choices to recover complementarity in utility and the demand system required for the policy counterfactuals.

IV Model and Identification

A The Model

I develop a discrete choice model of residential photovoltaic system and electric vehicle adoption following Berry, Levinsohn and Pakes (1995) and Gentzkow (2007).

The decision-maker is an individual. Each individual $i = 1, \dots, N$ in market $m = 1, \dots, M$ maximizes utility by choosing a single product j or a bundle of two products $\mathbf{b}$. I focus on bundles comprising two product categories indexed by C (car) and S (solar panel). The car is the individual's own vehicle; the solar panel is the system installed at the individual's residence. Let J_m^C denote the set of available car products in market m , and let J_m^S denote the set of solar panel choices in market m , where $J_m^S = \{0, 1\}$ represents the decision to either not purchase (0) or purchase (1) a PV system.

Assume that the bundles are ordered so that $\mathbf{b} = 0$ is the outside option, which consists of purchasing neither product (or choosing another product outside these categories). The indirect utility of individual i in market m from choosing a singleton product j (i.e., a product from either J_m^C or a PV system) is given by:

$$u_{imj} = \delta_{mj} + \mu_{imj} + \varepsilon_{imj} \quad (2)$$

$$u_{im0} = \varepsilon_{im0} \quad (3)$$

where δ_{mj} is the market m -specific average utility of product j , μ_{imj} is an individual-specific utility deviation from δ_{mj} , and ε_{imj} and ε_{im0} are error terms. The mean utility of singleton j can be further decomposed into observable and unobservable components as:

$$\delta_{mj} = \mathbf{x}_{mj}\boldsymbol{\beta} + \alpha p_{mj} + \xi_{mj} \quad (4)$$

where $\mathbf{x}_{mj}$ is a vector of observed product characteristics, $\boldsymbol{\beta}$ is the corresponding vector of coefficients, p_{mj} is the price of product j in market m , and ξ_{mj} captures unobserved product characteristics (which are observed by consumers and firms, but not by the researcher).

The individual-specific deviation from mean utility is defined by an interaction between ob-

servable consumer demographics and product characteristics as follows:

$$\mu_{imj} = \mathbf{X}'_{mj} \Phi \mathbf{D}_{im} \quad (5)$$

where $\mathbf{X}_{mj}$ is a vector of product characteristics (including price), $\mathbf{D}_{im}$ is a vector of observed characteristics of individual i and of the household to which i belongs—the individual’s own education, together with the household’s income, number of members, and age of its head, and the solar irradiation and public charging-station density of the household’s location—and Φ is a matrix of interaction coefficients.

To simplify exposition, I will refer to the products that bundle $\mathbf{b}$ contains as $j \in \mathbf{b}$. Following Gentzkow (2007), the indirect utility of individual i in market m from purchasing bundle $\mathbf{b}$ is:

$$\begin{aligned} U_{im\mathbf{b}} &= \sum_{j \in \mathbf{b}} (\delta_{mj} + \mu_{imj}) + \Gamma_{m\mathbf{b}} + \varepsilon_{im\mathbf{b}} \\ &= \sum_{j \in \mathbf{b}} \delta_{mj} + \Gamma_{m\mathbf{b}} + \sum_{j \in \mathbf{b}} \mu_{imj} + \varepsilon_{im\mathbf{b}} \\ &= \delta_{m\mathbf{b}} + \mu_{im\mathbf{b}} + \varepsilon_{im\mathbf{b}} \end{aligned} \quad (6)$$

$\Gamma_{m\mathbf{b}}$ is an average interaction between the products in a bundle $\mathbf{b}$ in market m that determines whether individuals, on average, obtain higher or lower utility from joint consumption. $\Gamma_{m\mathbf{b}}$ is set to 0 for singleton bundles. $\delta_{m\mathbf{b}} = \sum_{j \in \mathbf{b}} \delta_{mj} + \Gamma_{m\mathbf{b}}$ is the market m -specific average utility for bundle $\mathbf{b}$, $\mu_{im\mathbf{b}}$ is an individual-specific deviation from $\delta_{m\mathbf{b}}$, and $\varepsilon_{im\mathbf{b}}$ is an idiosyncratic taste shock, i.i.d. across (i, m, b) and assumed to be drawn from a type-1 extreme value distribution.¹⁵

B Identification

The parameters that have to be identified are the vector of mean product utilities (δ), coefficients on interactions between observable demographics and product characteristics Φ , coefficients on product characteristics β , the α -price coefficient, and the vector of interaction terms Γ .

¹⁵To address potential concerns about high upfront costs of the technologies, the model incorporates heterogeneity in budget constraints by interacting price with household income, ensuring that consumers with different income levels exhibit varying price sensitivities. This specification mirrors the approach used in Berry, Levinsohn and Pakes (1995) to account for budget constraints in differentiated product demand models. Additionally, similar to Bayer, Ferreira and McMillan (2007) and Bayer et al. (2016), who model housing demand, this approach acknowledges that lower-income consumers face greater financial constraints, making them less likely to buy expensive products. This method is commonly used in structural demand estimation, particularly in contexts where upfront costs and budget constraints significantly influence consumer choices.

1 Mean Utility and Consumer Preferences

All mean utilities (δ 's) are identified from within-market variation in product shares. The δ 's are uniquely determined so that the model-predicted average choice probabilities match the observed market shares.

Similarly, the coefficients on interactions between observable demographics and product characteristics, Φ , are identified by the observed covariances between the demographic characteristics of consumers that chose product j and the product's characteristics.¹⁶ In other words, Φ sets the model's prediction for the covariance between each demographic variable and a product characteristic equal to its population counterpart. Finally, coefficients on product characteristics (β) are identified by matching predicted and observed levels of product characteristics within and across markets.

2 Marginal Utility of Income

Instruments are generally required to identify α . This is because prices and ξ_{mj} will be correlated if unobserved product characteristics that increase consumer utility are capitalized into market prices. I do not observe all of the product attributes that may affect price. For example, for cars, I do not observe direct measures of comfort, design, ride smoothness, predicted resale value, prestige, etc.¹⁷

To deal with endogenous prices, I use BLP-style instruments based on the sum of observable characteristics of rival products within each market. The product characteristics included in the instrument set are horsepower, miles per gallon, and fuel type indicators (gasoline, hybrid, or electric vehicle). These instruments are relevant because the degree of differentiation between a product and its rivals affects the markup a firm can charge. Their validity rests on the assumption that observed product characteristics are determined independently of the unobserved demand shock ξ_{mj} . Related instruments are used in studies of EV demand by Xing, Leard and Li (2021), Springel (2021), and Li et al. (2021). Other studies use cost shifters or subsidy variation to identify demand elasticities, including Bollinger et al. (2023) for solar and storage and Li (2023) for EVs.

¹⁶Using the first-order conditions with respect to Φ , one can show that the maximum likelihood estimates of Φ are the ones that equate the observed and predicted covariance between the product's attributes and the demographic factors of the customers who selected that product. In the limit, as the number of consumers approaches infinity, Φ is identified as the solution to the system of the $L(K+1)$ equations, where L is the number of demographic variables and K is the number of product characteristics:

$$E_{\text{Population}} [x^k D^l] = E_{\text{Model}} [x^k D^l; \Phi]$$

¹⁷Moreover, prices might be linked to marketing initiatives. For example, a product may be promoted via advertising and special offers. The price will then be negatively correlated with advertising, with higher advertising accompanied by lower prices. As an alternative, businesses might charge more for their goods to cover the cost of advertising, which leads to a positive correlation. In either scenario, the product's pricing is no longer independent of the unobserved variables influencing consumers' purchasing decisions.

Appendix Section D.3 re-estimates the model using alternative instruments, including construction wages, input costs, and PV rebates, and finds that the estimated complementarity is stable across specifications.

3 The Interaction Term

The challenge with identifying Γ is distinguishing complementarity from unobserved preferences. In principle, there could be unobserved sources of heterogeneity not present in my model that could be conflated with complementarity. Therefore, I need a source of variation in the data that would reflect complementarity. Similar to Gentzkow (2007), I obtain this variation, in part, from between-market variation in “control variables” that affect the utility of one good but not the other. The utility of EVs, but not PVs, will be impacted by factors like average fuel economy, horsepower, and the number of public EV charging stations in the county where the household is located.¹⁸ Similarly, the county-level global horizontal irradiation rate is a factor that affects the utility of PVs but not EVs. Additionally, price variation also helps to identify complementarity terms.

Stepping back, Gentzkow (2007) shows that the sign of Γ_b determines whether the two products are substitutes or complements.¹⁹ If $\Gamma_b > 0$, products in the bundle are Hicksian complements with negative cross-price derivatives.²⁰ If $\Gamma_b < 0$, they are substitutes. If $\Gamma_b = 0$, the demand for one product is independent of the prices of another product in a different category.

To illustrate this, consider a simple model with two goods, A and B , and homogeneous consumers. Assume that utilities are given by:

$$u_{0m} = \varepsilon_{0m}, \quad (7)$$

$$u_{Am} = \delta_{Am} + \alpha p_{Am} + \varepsilon_{Am}, \quad (8)$$

$$u_{Bm} = \delta_{Bm} + \alpha p_{Bm} + \varepsilon_{Bm}, \quad (9)$$

$$u_{ABm} = \delta_{Am} + \delta_{Bm} + \alpha (p_{Am} + p_{Bm}) + \Gamma_{ABm} + \varepsilon_{ABm}, \quad (10)$$

where δ_{Am} and δ_{Bm} represent mean utilities in market m , and ε_{0m} , ε_{Am} , ε_{Bm} and ε_{ABm} are error terms.

Let P_A , P_B , P_{AB} denote probabilities of purchasing products A , B and bundle of A and B ,

¹⁸The instrument assumes that public EV charging infrastructure influences EV adoption but does not directly affect the utility of solar adoption. While it may indirectly raise electricity demand, it does not independently alter the appeal of rooftop solar. Moreover, EV owners are generally assumed to prefer home charging due to its cost advantages, making it unlikely that public charging availability substantially displaces residential electricity demand.

¹⁹See the discussion of Figure 1 in Gentzkow (2007) for an intuitive visualization.

²⁰Drawing on the lemma from Schlee and Khan (2022), when the nonnegativity constraint on the numeraire good is non-binding — meaning that consumer expenditures on goods like solar panels and electric vehicles do not fully exhaust their budget — Marshallian and Hicksian demands coincide, even with discrete goods. Consequently, the goods in question can be regarded as either Marshallian or Hicksian complements.

respectively. The total demand per consumer for good A is $Q_A = P_A + P_{AB}$, and for good B is $Q_B = P_B + P_{AB}$. Using the assumed parametric form for utility, the change in demand for good A with a one unit change in the price of good B can be expressed as follows:

$$\frac{dQ_{Am}}{dp_{Bm}} = -\alpha \cdot \frac{\exp(V_{Am} + V_{Bm}) - \exp(V_{Am} + V_{Bm} + \Gamma_{ABm})}{(1 + \exp(V_{Am}) + \exp(V_{Bm}) + \exp(V_{Am} + V_{Bm} + \Gamma_{ABm}))^2}, \quad (11)$$

where V_{jm} denotes the deterministic part of product j 's utility.

In equation 11, the denominator is strictly positive because it is squared, so the sign of the derivative depends solely on the numerator. Because higher prices lower utility, $\alpha < 0$, so $-\alpha > 0$. When $\Gamma_{AB} > 0$, the numerator is negative, indicating that an increase in p_B reduces demand for good A ; hence, the goods are complements. When $\Gamma_{AB} < 0$, the sign reverses, and the goods are substitutes. Finally, when $\Gamma_{AB} = 0$, the derivative is zero, and the goods are independent.

To clarify how Γ is identified, consider the four-alternative example above: neither good, A alone, B alone, and the bundle AB . A variable that increases the utility of owning A , without directly affecting the standalone utility of B or Γ , makes both A alone and AB more attractive. By the logit's IIA property, the shares of these two alternatives rise by the same proportion, while the shares of B alone and the outside option fall by the same proportion.

What distinguishes complements from substitutes is the response of total ownership of B , including joint ownership. When $\Gamma = 0$, the increase in the bundle share exactly offsets the decline in ownership of B alone, leaving total ownership of B unchanged. When $\Gamma > 0$, the increase in joint ownership exceeds that decline, so total ownership of B rises as A becomes more attractive. When $\Gamma < 0$, total ownership of B falls. Thus, Γ governs the cross-technology ownership response while preserving IIA across the four alternatives.

Empirically, technology-specific utility shifters and price variation help identify these cross-technology responses under the maintained exclusion restrictions, while demographic interactions account for observed differences in household preferences.

V Estimation

The model is static and is defined over ownership states rather than adoption events. Each household's observed technology portfolio at the survey date is mapped into one alternative in the choice set. Thus, a household observed owning both an EV and a PV system is assigned to the corresponding EV–PV bundle, regardless of whether the technologies were acquired simultaneously or sequentially. Because the order of acquisition is not observed, the model does not distinguish among different paths to the same ownership state. Accordingly, the complementarity parameter Γ

captures the additional utility associated with joint ownership rather than a preference specifically for simultaneous adoption.

The choice set consists of owning neither product, a good from one set only, a good from the other set only, or a bundle. The options in the car set include a gasoline car, a hybrid car, an EV, or none of the above. The household can choose among five sizes for gas cars (compact, midsize, SUV, pick-up, van), three sizes for hybrids (compact, midsize, SUV), and two sizes for EVs (compact and midsize or bigger). I consolidated the midsize EV and SUV EV categories to minimize the number of products with zero observed market share, given the limited availability of SUV options in 2013.²¹ The solar panel set consists of either installing a solar panel or not. Thus, there are 22 possible alternatives, including the outside option. The outside option corresponds to owning neither a car in the modeled choice set nor a solar panel.²²

The dataset is structured at the individual level, with some individuals associated with more than one vehicle, which could introduce some double counting. In cases where the primary driver of a vehicle is not identified, the vehicle is assigned to the household head. Notably, only 8% of individuals in the sample are associated with driving more than one vehicle. Finally, I exclude sports cars, diesel, hydrogen, plug-in hydrogen, gasoline-ethanol flex fuel, and compressed natural gas vehicles from the sample. This exclusion is due to the limited number of observations for these cars and their substantial differences from the main categories, making it impractical to categorize them appropriately. In total, there are 19,859 individual-car level observations.

The vector of product characteristics $\mathbf{x}_{mj}$ includes miles per gallon, horsepower, the vehicle's length multiplied by its width and by its height, dollars per mile (the cost of driving a car), and ownership of PV.²³ The vector of observable household characteristics, D_{im} , includes dummy variables for income groups and for the age of the household head, the number of household members, and an indicator for whether the individual has a bachelor's degree or higher level of education. I divide households into four annual income groups: 1) below \$50,000; 2) between \$50,000 and \$100,000; 3) between \$100,000 and \$200,000; 4) above \$200,000. Finally, the number

²¹According to the U.S. Department of Energy, in 2013, there was just one model of SUV electric car available (Toyota RAV4).

²²Households choose among gas, hybrid, and electric vehicles; solar panels; vehicle-solar bundles; and the outside option of adopting neither technology. A low taste for green technology is therefore reflected in a low predicted probability of choosing a PV or EV alternative. Heterogeneity in these preferences by income, education, age, and household size is captured through interactions between product characteristics and household demographics. I assume that all alternatives are available for consideration by all households and do not separately model a latent consideration set.

²³For gasoline vehicles, I compute the operating cost per mile by dividing the average annual retail gasoline price in California by the vehicle's fuel economy (miles per gallon). For electric vehicles, I use the average residential electricity price in California for the corresponding year together with the vehicle's MPGe rating. Following the EPA convention, one gallon of gasoline is treated as energy-equivalent to 33.7 kilowatt-hours of electricity. This allows me to convert efficiency into kilowatt-hours per mile, given by 33.7/MPGe, and then multiply by the electricity price to obtain the cost per mile.

of public charging stations in the household’s residential county enters the utility of the products that contain electric vehicles, while the global horizontal irradiation rate in the county enters the utility of products with solar panels.²⁴

The data impose two limitations on the analysis. First, limited information on the specific products owned motivates grouping vehicles by size and fuel type. Second, the repeated cross-sections do not track household ownership transitions, and PV installation dates are unavailable. A fully dynamic adoption model would therefore require additional assumptions about unobserved adoption histories and expectations. Appendix E uses a two-period extension to assess how forward-looking adoption affects the interpretation of the complementarity estimate and the policy conclusions.

Estimation proceeds in two stages, following Berry, Levinsohn and Pakes (1995) and Bayer, Ferreira and McMillan (2007). In the first stage, I estimate the heterogeneous preference parameters, Φ , by maximum likelihood and recover the product-market mean utilities, δ , using a contraction mapping. I recover a separate mean utility for each product in each market, allowing mean utilities to vary flexibly across regions and survey years.²⁵

In the second stage, I decompose the recovered mean utilities into observed and unobserved characteristics according to equation 4. Because price may be correlated with unobserved product characteristics, I instrument for price using the BLP-style sums of characteristics of rival products within the market, as described in Section IV.B.2. As a robustness check, I replace these instruments with alternative supply-side cost shifters; Appendix Section D.3 provides details. I compute asymptotic standard errors from the first-stage observed information matrix and adjust the second-stage IV standard errors for estimation error in the recovered mean utilities, following Newey and McFadden (1994).

VI Results

A Parameter Estimates

Table 1 presents the model estimates.²⁶ The left side displays the estimated coefficients for product characteristics. Most coefficients are precisely estimated. The average household has a positive

²⁴California’s HOV-lane access for qualifying electric vehicles is a further potential shifter of EV utility, and I collected the number of high-occupancy vehicle lanes by region for this purpose. I tested a specification that includes the number of HOV lanes in the region as a product characteristic in the second stage. Its coefficient is statistically insignificant and its inclusion does not affect the remaining estimates, so I exclude it from the final specification.

²⁵Due to the small number of observations in some markets, I replace zero observed shares by assuming that one individual in the market chooses the corresponding product, which allows me to implement the contraction mapping.

²⁶Table B.5 in the appendix shows the quality of model fit. I estimate model-predicted shares for the whole population and then product shares within different income group households. The table shows that the model does a good job of matching the observed shares in the data.

valuation of vehicle horsepower, miles per gallon, solar panels, and synergy between EVs and PVs. I evaluate the utility of mileage per gallon separately, conditional on purchasing an EV and the other types of cars. We can see that for gas and hybrid cars, the coefficient on mileage per gallon is positive. This implies that consumers get positive utility from higher fuel efficiency. However, the negative coefficient on miles per gallon equivalent (MPGe) for electric vehicles might be picking up some unobserved characteristics that correlate with MPGe, such as luxury features and performance packages that drain the battery. Moreover, households experience a decrease in utility on average due to higher prices and bigger sizes of cars.

The right side of Table 1 displays the estimates of the coefficients on interactions of observable demographic characteristics with product characteristics. Most coefficients have the expected signs. Higher-income households exhibit lower sensitivity to vehicle prices. Furthermore, these households place greater value on the reduced operational costs associated with driving the vehicle. This finding is intuitive as wealthier households have a greater capacity to make upfront investments that lead to long-term cost savings. Essentially, higher-income households are more likely to prioritize and invest in vehicles with lower running costs, reflecting their ability to absorb higher initial expenditures for future financial benefits. Larger families tend to place less value on larger vehicles, likely because they opt for multiple smaller, more affordable cars rather than a single, more expensive large vehicle. In addition, we can see that more highly educated consumers value fuel efficiency and solar systems more than people with lower levels of education. In addition, the estimates indicate that older individuals place a higher value on solar panels. This may be attributable to their higher accumulated wealth, which provides them with the financial flexibility to invest in solar panel installations. Lastly, households residing in counties with higher numbers of public EV charging stations get higher utility from EVs, whereas households living in places with higher global horizontal irradiation rates value solar panels more.

The results are robust to alternative specifications. The estimates are similar when I account for potentially endogenous charging-station placement using a control-function approach (Appendix Section D.1), when I allow for a common unobserved individual-specific taste for EVs and PVs (Appendix Section D.2), and when I replace the BLP-style price instruments with supply-side cost shifters (Appendix Section D.3). Repeating the policy counterfactuals separately by survey cohort yields similar conclusions (Appendix Section D.4).²⁷

²⁷As a placebo check, I compute the excess of each gasoline–PV bundle’s mean utility over the sum of its components’ mean utilities, using the freely estimated product-market mean utilities. These estimates range from -0.05 to 0.16 across gasoline vehicle categories, compared with 1.54 – 1.67 for EVs, suggesting that the strong EV–PV interaction does not extend to vehicle ownership generally.

Table 1: Parameter Estimates of Mean Utility and First Stage Parameters

Mean Utility		First Stage Parameters	
Constant	-0.160 (0.071)	Income (50-100k) $\times$ prices	0.223 (0.023)
MPG_{EV}	-2.032 (0.182)	Income (50-100k) $\times$ \$ per mile	-1.587 (0.638)
MPGgas/hybrid	1.095 (0.288)	Income (100-200k) $\times$ prices	0.372 (0.023)
Horsepower	2.984 (0.109)	Income (100-200k) $\times$ \$ per mile	-3.086 (0.633)
Car size	-0.407 (0.032)	Income (200k or more) $\times$ prices	0.521 (0.024)
PV	0.646 (0.629)	Income (200k or more) $\times$ \$ per mile	-5.515 (0.717)
Γ	1.366 (0.160)	HH members $\times$ Car size	-0.043 (0.003)
Prices	-1.641 (0.059)	College degree $\times$ mpg	2.213 (0.104)
		College degree $\times$ PV	0.041 (0.043)
		GHI $\times$ PV	2.519 (1.137)
		Stations $\times$ EV	0.199 (0.157)
		Age (35-64) $\times$ PV	-0.088 (0.067)
		Age (35-64) $\times$ EV	0.208 (0.130)
		Age (65 or older) $\times$ PV	0.285 (0.072)
		Age (65 or older) $\times$ EV	-0.031 (0.142)
Observations	396		19,859

Notes: Standard errors are reported in parentheses. For the first-stage preference parameters, standard errors are computed from the inverse observed information matrix for the joint estimation of preference parameters and product mean utilities. Standard errors for the second-stage IV estimates account for uncertainty in the estimated mean utilities, following the two-step variance approach of Newey and McFadden (1994).

B Results Interpretation

Table 2 presents estimates of marginal willingness to pay (MWTP) to help interpret parameter differences between income groups. The results show that higher-income households are willing

to pay more for higher fuel efficiency and horsepower. For example, households with an annual income above \$200,000 are willing to pay \$267 for an additional unit of horsepower. In comparison, households with less than \$50,000 per year are willing to pay \$182 for an extra unit of horsepower. The willingness to pay for PV also increases with income, ranging from \$12,494 for lower-income households to \$17,813 for the highest-income households.

The WTP estimates for the synergy between EVs and PVs range from \$7,877 to \$11,859, increasing with household income.²⁸ These amounts represent the additional willingness to pay for an EV associated with owning a PV system, holding other household characteristics fixed. For example, a household earning less than \$50,000 annually would be willing to pay \$8,056 more for a compact EV with PV ownership than without it; the corresponding premium for a household earning more than \$200,000 is \$10,895. As the choice set distinguishes compact EV and PV bundles from larger EV and PV bundles, the implied WTP naturally varies by vehicle size. The differences across vehicle size are small relative to those across income groups and are not uniform in sign: the premium for the larger EV exceeds that for the compact EV in the three higher income groups but not in the lowest.

On average, a 1% permanent decrease in the price of EVs is associated with a 0.23% increase in the long-run ownership rate of PV systems; similarly, a 1% permanent decrease in the price of PVs is associated with a 0.69% increase in the long-run ownership rate of EVs.²⁹ Applied locally, a 10% permanent price reduction implies an increase of about 0.3 percentage points in PV ownership (from a baseline ownership rate of 14.5%) and about 0.2 percentage points in EV ownership (from a baseline of 2.4%). The larger elasticity of EV ownership largely reflects its smaller baseline share: because the EV ownership base is much smaller than the PV base, a given percentage-point response translates into a several-fold larger proportional change for EVs. The asymmetry also could reflect substitution possibilities within the vehicle market: households facing higher EV prices can switch to gasoline or hybrid vehicles while retaining their PV systems, whereas households facing higher PV prices have no close substitute for solar within the choice set.

C Complementarity Decomposition

I decompose complementarity into two primary channels: electricity cost savings and consumer preferences. A portion of this complementarity arises from the potential cost savings associated with the combined use of EVs and PVs. The remaining portion is attributed to consumers' willingness

²⁸These premiums are calculated from market- and vehicle-specific complementarity measures recovered as the difference between each bundle's mean utility and the sum of its components' mean utilities, divided by the absolute value of the income-specific price coefficient and converted to dollars. They therefore vary by vehicle size and are distinct from a direct monetary conversion of the pooled complementarity coefficient reported in Table 1.

²⁹Because the model is defined over ownership states, these are long-run stock elasticities rather than elasticities of annual sales or installation flows.

Table 2: Willingness to Pay Estimates, US dollars

	(I)	(II)	(III)	(IV)
Income	≤ 50 k	50-100 k	100-200 k	≥ 200 k
MPG	84	118	157	210
Horsepower	182	211	235	267
Car Size	-33	-38	-43	-48
PV	12,494	14,490	15,947	17,813
EV compact+PV Interaction	8,056	9,325	10,536	10,895
EV midsize+PV Interaction	7,877	9,476	11,079	11,859

Notes: Each entry is the marginal utility of the attribute divided by the marginal utility of income, evaluated at the household level and averaged within income group using population weights. For attributes that interact with demographics, the numerator is the mean-utility coefficient plus the income group's average interaction.

to adopt environmentally friendly practices. Households may benefit from the combined ownership of PV systems and EVs, as installing solar panels can reduce electricity costs by shifting increased consumption from EVs to lower price tiers. In addition, EV time-of-use (TOU) rates that require having an EV allow households to sell excess solar energy at higher rates during peak generation hours.

I use information on residential location to link individuals to their utility companies, including Southern California Edison (SCE), San Diego Gas & Electric (SDG&E), and Pacific Gas and Electric Company (PG&E).³⁰ Then, I use utility-specific information on rate plans to calculate the average savings that consumers may realize by comparing electric bills under different rate plans and discounting the results at 3% annually over ten years. Additional details on these calculations are provided in Appendix G.³¹

On average, the discounted complementarity savings over ten years amount to approximately \$1,775 for customers on the block rate plan, and \$2,081 for those who choose the cost-minimizing plan. The model-predicted WTP for complementarity varies significantly across markets, with the highest values observed in 2013, followed by a decline in most markets by 2017 and 2019.³² This pattern likely reflects the early adopters' strong preference for environmentally friendly goods. Appendix D.4 repeats the policy counterfactuals separately for the 2013 and 2017–2019 markets.

³⁰Additionally, the dataset includes a small percentage of individuals serviced by Sacramento Municipal Utility District, Pacific Power, Liberty Utilities, Lassen Municipal Utility District, and Imperial Irrigation District. Due to their minimal representation, I exclude these companies from the calculation of the average savings for each region.

³¹Table C.6 presents the cost savings due to complementarity if a customer is on the block rate plan and if they are on the cost-minimizing plan, which could either be the block rate plan or one of the time-of-use plans offered by utility companies, based on the assumed consumption profile. Although the TOU plan may provide greater savings, most customers continue to use the block rate plan. The table also presents the model's predicted willingness to pay (WTP) for complementarity across markets.

³²This variation arises because the complementarity term can be calculated separately for each market, based on the difference between the utility of adopting both technologies together and the sum of the utilities of adopting each one individually. That market-specific complementarity measure is divided by each household's own price coefficient, then averaged across households within the market using population weights.

Although the magnitude of complementarity and the policy effects differ across cohorts, the benefit-maximizing subsidy allocation is similar in both. The mean model-predicted WTP across all markets is \$10,223, and \$7,073 when 2013 is excluded. Over the same 2017–2019 markets, savings on the cost-minimizing plan average \$2,390, so modeled bill savings account for around 34% of the WTP for complementarity, while the remainder reflects consumers’ valuation of additional benefits, such as their preference for environmentally friendly choices.³³

As adoption grows over time, marginal adopters may attach less value to environmental benefits, which would diminish the strength of complementarities. From a policy perspective, because roughly one-third of complementarity is attributable to bill savings, favorable TOU tariffs and net-energy-metering rules are important but they can only influence the savings-driven component. Stimulating additional demand likely requires non-price levers, most notably, information and outreach that emphasize the environmental benefits of EVs and PVs.³⁴

Model Scope and Extensions Under the maintained specification, complementarity is common across households within a market and therefore represents the market-level average additional utility from joint ownership. The estimate draws on choices across the full sample, including adopters and nonadopters. This average interpretation relies on the homogeneity assumption. If complementarity differs across households, the common estimate need not equal their arithmetic mean: because ownership probabilities respond nonlinearly to complementarity, matching observed ownership patterns is not equivalent to averaging individual valuations. Allowing for a shared unobserved taste for green technologies leaves the estimate largely unchanged (Appendix Section D.2), suggesting that the finding does not simply reflect environmentally oriented households selecting into ownership of both technologies. A natural extension would allow households to differ also in the additional value they derive from joint ownership. This would allow subsidy counterfactuals to account for differences in complementarity between existing and newly induced

³³Some manufacturers engaged in joint marketing and offered cross-technology incentives during the sample period. Ford’s “Drive Green for Life” program offered a \$750 rebate on a SunPower residential solar system (Ford Motor Company, 2017), and BMW announced a 10% SolarCity discount for BMW i owners available from early 2014 (BMW Group, 2013). Honda and Acura buyers in SolarCity’s service territory, including California, were also offered a \$400 residential solar discount, although eligibility was not restricted to EV buyers (Renewable Energy World, 2013). Conditional purchase discounts provide additional financial benefits of joint ownership, alongside the operating-cost savings discussed above, while joint marketing may increase awareness of these benefits. The fixed-dollar rebates offered by these companies were modest relative to the estimated willingness to pay for complementarity, although this comparison does not quantify the contribution of joint marketing. To the extent these incentives are not reflected in observed prices, they may contribute to the estimated complementarity but are not separately measured in the calculation of operating-cost savings. Identifying their contribution, and that of joint marketing, would require data on program participation and marketing exposure.

³⁴The decomposition does not limit the purchase-subsidy counterfactuals in Section VII: those subsidies change purchase prices, and the model uses the full estimated Γ to predict adoption. The split matters when evaluating policies that affect only one source of complementarity. For example, applying the full Γ to a rate-design reform that changes operating costs would overstate its effect on joint adoption.

adopters, clarifying how adoption responses change as subsidies expand.

The model accommodates changes in prices and product characteristics and recovers separate mean utilities for each region–year market. The complementarity measures recovered from these mean utilities can likewise vary across regions and survey years. The coefficients on product characteristics and their interactions with household demographics, however, are held fixed across survey years. Allowing these coefficients to evolve would accommodate changes in price sensitivity and the valuation of product attributes as consumers become more familiar with the technologies. Together with household-level heterogeneity in complementarity, this extension would provide a framework for studying how evolving preferences and changes in adopter composition shape subsidy effectiveness as these technologies diffuse.

VII Counterfactual and Policy Experiments

I perform several counterfactual experiments using the estimated model. Specifically, I consider (i) the importance of complementarity in the adoption of technologies; (ii) the magnitude of the subsidy effects on the adoption of technology with and without accounting for complementarity between goods; and (iii) the cost-minimizing design of subsidies between the two markets.

A Quantifying the Complementarity Effect

In the first column of Table 3, we can see the predicted shares of each product (which, by construction, perfectly match the observed shares) in the model that allows for flexible substitution patterns among the goods. The second column presents the shares that would be observed if EVs and PVs were restricted to be independent, given current market prices and subsidies. The results imply that the share of bundles that include solar panels and electric vehicles would fall by 64%, and in total consumers would buy fewer EVs and PVs.

Table 3: Product shares, %

Product	<i>Model predictions</i>		
	w/ complementarity	Independent goods	% Change
All Solar	14.5	14.0	-3.6
All EV	2.4	1.7	-26.6
PV & EV bundle	1.0	0.4	-64.0
Only PV	13.5	13.6	1.0
Only EV	1.3	1.4	2.4

B Complementarity Role in Subsidy Effects

The model also enables me to conduct back-of-the-envelope calculations of the effects of subsidies on carbon emissions and particulate matter ($PM_{2.5}$) reductions. For the analysis, I rely on data from several government sources and research reports to make assumptions. Key assumptions include California's potential market size of 20.7 million people and a typical residential solar system generating 9,000 kWh per year. Additionally, electricity generation emissions are estimated to be 457.5 *lb/MWh* for CO_2 and 0.028 *lb/MWh* for $PM_{2.5}$. Vehicle CO_2 emissions vary by fuel type, with gasoline cars emitting 12,594 lbs CO_2 annually. For $PM_{2.5}$, I assume that gas cars emit at the current emission standard of 0.003 grams per mile. Finally, the average annual vehicle miles traveled (VMT) is 14,489 miles. Supporting details are provided in Appendix H.

Next, I assume that the California government has \$2 billion to spend on EV and PV subsidies. This amount has been selected as it approximates the current expenditure by California on the promotion of these technologies.³⁵ I compute the predicted CO_2 reductions when the budget is spent on solar panels alone and then on EVs alone. Treating the two technologies as independent goods, the EV subsidy is the more effective of the two, cutting 1.06 million metric tons per year against 0.69 for PVs (Figure 3a). Two channels work against each other here. A subsidy dollar induces more PV adoption than EV adoption. However, one electric vehicle displaces considerably more CO_2 than one solar system at the margin of the California grid, and this per-unit advantage outweighs the adoption-per-dollar one. Allowing for complementarity raises both figures, by 0.43 and 0.31 million metric tons, respectively, for totals of 1.49 and 1.00. Accounting for the interdependence therefore allows a given emissions target to be met with a smaller budget.

The ranking reverses for $PM_{2.5}$ (Figure 3b), where the PV subsidy avoids 47 metric tons a year against 20 for the EV subsidy. Here the per-unit channel favors solar: one PV system avoids about 2.7 times as much $PM_{2.5}$ per year as replacing one gasoline vehicle with an EV. Complementarity again matters most for the EV subsidy, since about 43% of the $PM_{2.5}$ reductions it delivers come through induced solar adoption.

C Optimal Subsidy Allocation

Next, I analyze different ways to allocate subsidies between PVs and EVs to incentivize their adoption. I vary the share of the budget allocated to each technology continuously, from full support for PVs, to full support for EVs, to any combination of the two. Additionally, I consider subsidies for purchasing both technologies as a bundle. The results can advance our understanding

³⁵For example, in 2022, the Governor's budget included \$922.4 million over two years for the Equitable Building Decarbonization program, which supports residential electrification and includes PV system installation (Legislative Analyst's Office, 2022b). In addition, the state proposed \$6.1 billion over five years for various zero-emission vehicle (ZEV) initiatives, which translates to over \$1.2 billion annually (Legislative Analyst's Office, 2022a).

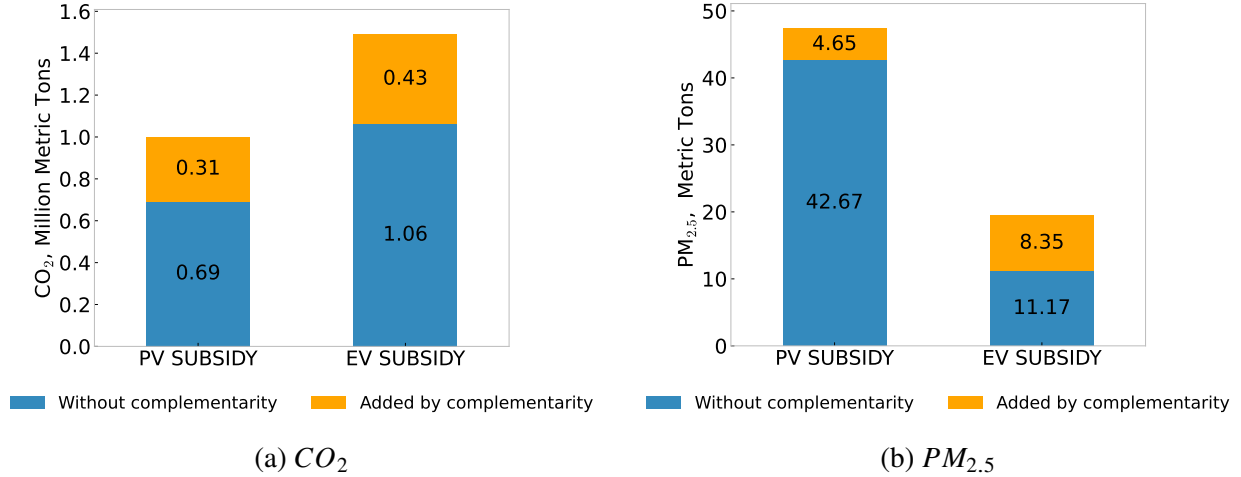


Figure 3: Emission reductions per year from EV and PV adoption

of the most cost-effective ways to reduce air pollutants.

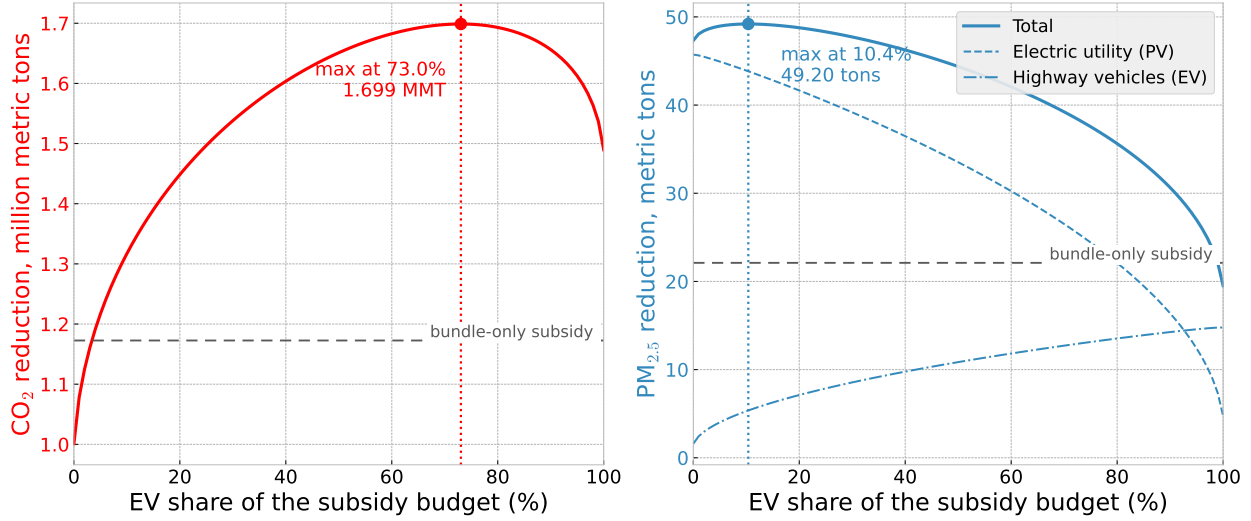
Figure 4a demonstrates the effects of subsidies on annual carbon dioxide and particulate matter emissions. The results of the full model imply that the most effective way to reduce CO_2 emissions is to allocate 73% of the budget to subsidies for electric vehicles and 27% to PVs. In contrast, the most significant decreases in $PM_{2.5}$ come from allocating most of the budget to solar systems: the reduction peaks at a 90% PV share and declines thereafter as more of the budget is shifted to EVs. Therefore, choosing which emissions target to prioritize will determine the most effective way to provide subsidies.

A subsidy restricted to joint purchases, shown as the dashed reference line in Figure 4a, reduces 1.17 million metric tons of CO_2 and 22.1 metric tons of $PM_{2.5}$ per year, less on both pollutants than most mixed allocations of the same budget. Restricting eligibility to households buying both technologies raises the cost per induced adopter and it pays for joint adoption that a single-technology subsidy already induces through the complementarity channel.

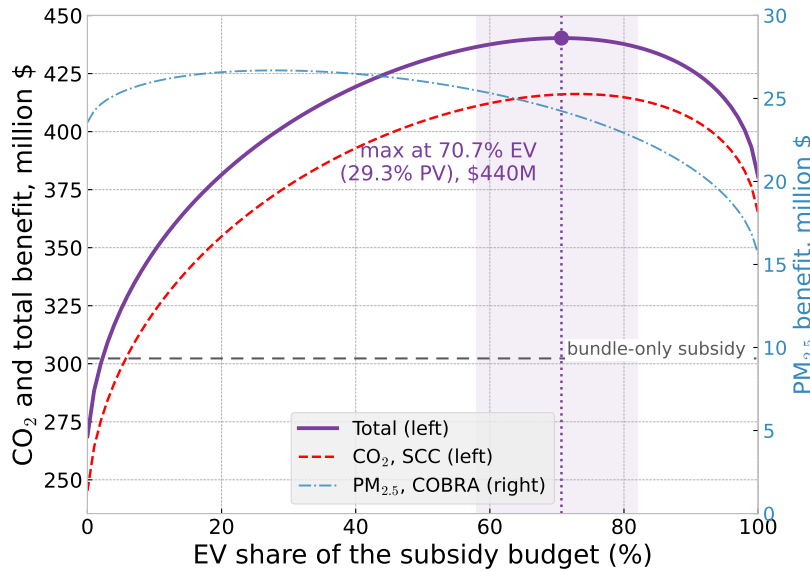
Moreover, I calculate the total annual monetary benefits in order to see which allocation is optimal in terms of investment returns. I apply the EPA's Social Cost of Carbon (SCC) for the emission year 2024, valued at \$245 per metric ton in 2024 dollars (U.S. Environmental Protection Agency, 2023).³⁶ At the benefit-maximizing allocation, reductions in CO_2 alone are worth approximately \$416 million per year. For $PM_{2.5}$, I use the EPA's CO-Benefits Risk Assessment Health Impacts Screening and Mapping Tool (COBRA; U.S. Environmental Protection Agency, 2024), which provides region-specific estimates of the economic value of health benefits from emission reductions in targeted sectors. To reflect the timing of health impacts, such as delayed changes in mortality and non-fatal heart attacks, COBRA applies a 2% discount rate, as recommended by the EPA. The results are presented in Figure 4b. Maximizing the combined monetized benefit

³⁶I adjust for inflation the value of 204\$ in 2020 dollars for the emission year 2024 using the consumer price index.

over the continuous budget share gives an optimum of 29% of resources to PV and 71% to EV, worth \$440 million per year. The objective is flat near this point: any split between 18% and 42% PV is within one percent of the maximum, so the policy recommendation is robust to missing the optimum by a wide margin. Allowing for forward-looking adoption shifts the optimum modestly toward EVs, and the static allocation remains within 0.4% of the best attainable benefit in every dynamic specification (Appendix Section E.2). Maximizing CO_2 reductions alone would move the optimum to 27% PV, and maximizing $PM_{2.5}$ emissions reductions alone to 90%, so the interior optimum reflects the relative size of the two damages. The payback period of the investment is about 5 years and 8 months at the discount rate of 7% and 4 years and 11 months at 3%.



(a) Annual Emissions Reductions



(b) Annual Total Monetary Value

Figure 4: Comparison of Annual Reductions and Benefits

Notes: Both panels vary the EV share of the fixed \$2 billion budget continuously. Panel (a) separates the two pollutants, and shows the $PM_{2.5}$ reduction decomposed into the electric-utility reduction that solar adoption delivers and the highway reduction that EV adoption delivers. Dotted vertical lines mark each maximizer. In panel (b), $PM_{2.5}$ is plotted on the right axis and accounts for \$24 million of the \$440 million at the optimum. The shaded band covers the allocations within one percent of the maximum. The dashed line in each panel is the bundle-only subsidy, which spends the same budget exclusively on joint purchases.

VIII Conclusion

This study contributes to the ongoing global discussion on green technology investment policy by providing a theoretically motivated analysis of the complementarity effects between two industries,

solar panels and electric vehicles. I presented descriptive evidence that the odds of installing solar panels are roughly three times higher for households with an EV than for those without one, and likewise, the odds of owning an EV are about three times higher for households with solar panels than for those without one. There are two potential explanations for this result. First, households with high utility from adopting solar panels also get high utility from purchasing an EV. Another reason is that the two goods complement each other.

I used a two-stage estimation procedure to recover structural parameters of the joint demand for solar systems and electric vehicles. These are the first estimates for substitution/complementarity patterns in the demand system for solar panels and electric vehicles. This is also the first application of a multi-product demand model to consider externalities. The results suggest that the two technologies are complements. The potential channels for complementarity are decreased costs and higher environmental benefits.

I implemented several counterfactual policy experiments. They show that complementarity between solar panels and electric vehicles plays a significant role in how subsidizing consumer purchases in either market affects air pollution. In addition, I analyzed the range of potential allocations of a fixed government budget on EV and PV subsidies and showed how this would reduce CO_2 and $PM_{2.5}$ emissions. The optimal allocation based on back-of-the-envelope calculations using conventional assumptions for monetizing damages from global warming and health effects of air pollution is to invest 71% of the resources into EV subsidies and 29% into PV subsidies.

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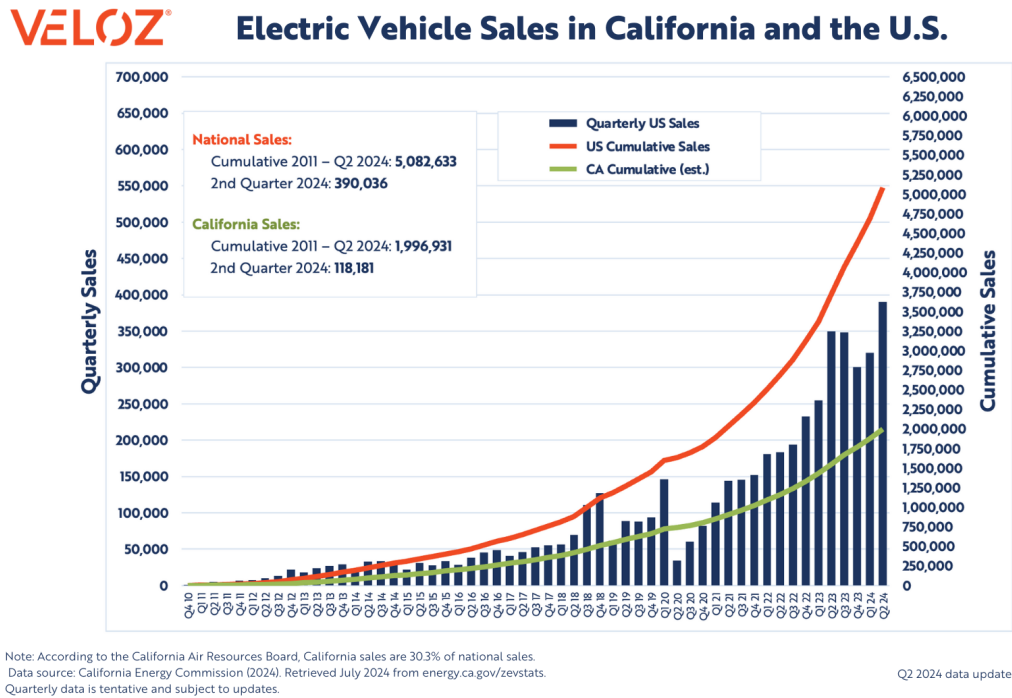
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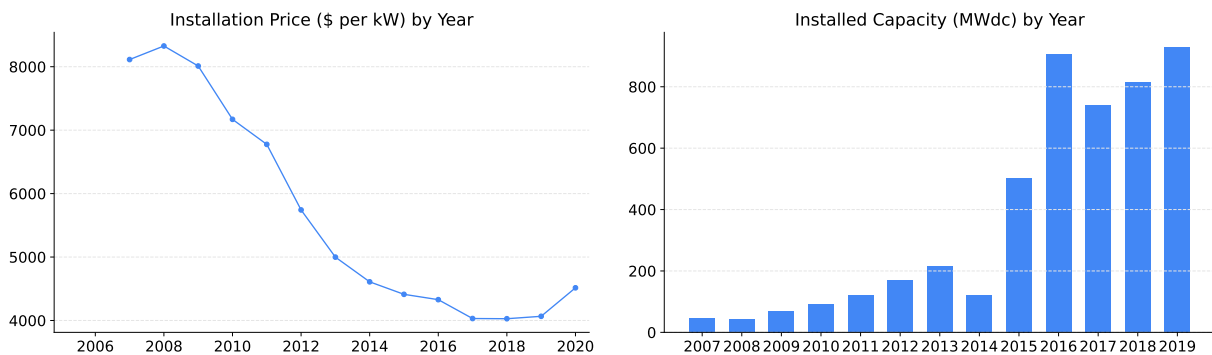
A Supplementary Figures

Figure A.1: Electric Vehicle Sales in California and the U.S.



Source: Veloz (2024)

Figure A.2: California Solar PV Pricing Trends & Deployment Growth



Source: Lawrence Berkeley National Laboratory (2023)

Table A.1: Summary Statistics

	California Vehicle Survey	California CPS
N of people	19,859	38,795
N of households	9,096	8,883
< \$50,000	16%	28%
\$50,000 - 99,999	32%	29%
\$100,000 - 199,999	37%	29%
> \$200,000	16%	13%
Black or African American	3%	5%
Hispanic	13%	33%
Female	49%	51%
Highest schooling:		
< High school	5%	14%
High school/GED	11%	24%
Associate's degree	12%	9%
Some College	16%	21%
College (4 year)	30%	21%
Post-graduate	27%	11%
Age group:		
18-34 years old	21%	28%
35-64 years old	59%	53%
65 or older	20%	20%

Note: CPS: IPUMS-CPS, Annual Social and Economic Supplement 2013, 2017 and 2019 (Flood et al., 2023). CPS counts are in thousands and are the average of the three years' weighted counts: all California residents, and households living in one-unit or mobile homes. CPS shares are weighted and pooled over the three years, for adults living in one-unit or mobile homes.

B Logistic Regression Results and Model Fit

Table B.2: Logistic Regression Results for EV Ownership

	(1)	(2)	(3)	(4)	(5)
Own PV	4.52 (0.329)	3.62 (0.270)	3.57 (0.268)	3.22 (0.252)	3.28 (0.257)
Income (25k-34k)		0.60 (0.380)	0.59 (0.374)	0.50 (0.318)	0.53 (0.334)
Income (35k-49k)		1.16 (0.538)	1.12 (0.520)	0.96 (0.451)	1.02 (0.480)
Income (50k-74k)		2.42 (0.971)	2.34 (0.941)	2.11 (0.852)	2.25 (0.912)
Income (75k-99k)		3.46 (1.362)	3.32 (1.306)	3.07 (1.217)	3.27 (1.297)
Income (100k-149k)		4.28 (1.659)	4.08 (1.582)	3.66 (1.429)	3.93 (1.537)
Income (150k-199k)		5.58 (2.181)	5.28 (2.065)	4.85 (1.915)	5.24 (2.073)
Income (200k-249k)		8.60 (3.377)	8.25 (3.243)	6.27 (2.489)	6.82 (2.712)
Income (250k+)		11.01 (4.279)	10.45 (4.065)	8.38 (3.304)	9.18 (3.628)
Age (35-64)			1.62 (0.185)	1.59 (0.190)	1.52 (0.182)
Age (65+)			1.83 (0.238)	1.28 (0.174)	1.10 (0.156)
HH size					0.88 (0.029)
County FE				✓	✓
Year FE				✓	✓
Observations	19,859	19,859	19,745	19,311	19,311

Note: Exponentiated coefficients. Standard errors are in parentheses.

Table B.3: Logistic regression results for PV Ownership

	(1)	(2)	(3)	(4)	(5)
Own EV	4.52 (0.329)	3.62 (0.270)	3.57 (0.268)	3.11 (0.240)	3.13 (0.242)
Income (25,000–34,999)		1.32 (0.253)	1.27 (0.245)	1.18 (0.230)	1.16 (0.226)
Income (35,000–49,999)		1.94 (0.318)	1.89 (0.312)	1.82 (0.303)	1.77 (0.295)
Income (50,000–74,999)		1.98 (0.301)	1.93 (0.293)	1.99 (0.307)	1.94 (0.300)
Income (75,000–99,999)		2.60 (0.387)	2.57 (0.383)	2.68 (0.406)	2.63 (0.398)
Income (100,000–149,999)		3.19 (0.465)	3.18 (0.464)	3.51 (0.521)	3.42 (0.508)
Income (150,000–199,999)		3.51 (0.525)	3.63 (0.544)	4.22 (0.645)	4.11 (0.629)
Income (200,000–249,999)		4.09 (0.632)	4.31 (0.668)	4.87 (0.771)	4.74 (0.752)
Income (250,000+)		5.95 (0.895)	6.27 (0.947)	7.70 (1.193)	7.46 (1.158)
Age (35–64)			0.89 (0.047)	0.94 (0.051)	0.97 (0.054)
Age (65+)			1.47 (0.089)	1.38 (0.088)	1.48 (0.100)
HH size					1.06 (0.017)
County FE				✓	✓
Year FE				✓	✓
Observations	19,859	19,859	19,745	19,595	19,595

Note: Exponentiated coefficients. Standard errors in parentheses.

Table B.4: Logit model results predicting PV and EV ownership.

	Dependent variable is EV ownership		Dependent variable is PV ownership	
	Marginal Effect	Standard Error	Marginal Effect	Standard Error
PV ownership	0.057	0.005		
EV ownership			0.193	0.016

Table B.5: Model Fit

Product	All observations		\$50,000-\$100,000		\$100,000-\$200,000		≥ \$200,000	
	Data	Model	Data	Model	Data	Model	Data	Model
None	0.107	0.107	0.119	0.116	0.079	0.081	0.049	0.059
PV	0.019	0.019	0.018	0.019	0.019	0.018	0.022	0.018
Gas compact	0.123	0.123	0.131	0.132	0.111	0.114	0.100	0.097
Gas midsize	0.229	0.229	0.245	0.235	0.233	0.235	0.191	0.211
Gas suv	0.183	0.183	0.182	0.190	0.198	0.187	0.187	0.171
Gas van	0.042	0.042	0.044	0.042	0.042	0.044	0.040	0.041
Gas pickup	0.077	0.077	0.088	0.084	0.071	0.075	0.043	0.057
Hyb compact	0.016	0.016	0.015	0.015	0.016	0.016	0.021	0.017
Hyb midsize	0.027	0.027	0.024	0.025	0.032	0.029	0.037	0.033
Hyb suv	0.008	0.008	0.005	0.007	0.009	0.008	0.014	0.011
EV compact	0.012	0.012	0.008	0.010	0.013	0.013	0.024	0.019
EV big	0.011	0.011	0.008	0.005	0.013	0.012	0.023	0.031
Gas comp+sol	0.020	0.020	0.017	0.019	0.022	0.022	0.029	0.024
Gas mid+sol	0.034	0.034	0.030	0.029	0.036	0.040	0.050	0.049
Gas suv+sol	0.034	0.034	0.025	0.029	0.042	0.039	0.056	0.050
Gas van+sol	0.007	0.007	0.006	0.006	0.008	0.009	0.014	0.012
Gas pickup+sol	0.014	0.014	0.014	0.013	0.015	0.016	0.015	0.017
Hyb comp+sol	0.006	0.006	0.004	0.005	0.008	0.007	0.010	0.009
Hyb mid+sol	0.009	0.009	0.006	0.006	0.010	0.010	0.017	0.016
Hyb suv+sol	0.003	0.003	0.002	0.002	0.003	0.003	0.006	0.005
EV comp+sol	0.009	0.009	0.005	0.006	0.010	0.011	0.024	0.021
EV big+sol	0.010	0.010	0.005	0.003	0.009	0.010	0.027	0.032
Observations	19,859		6,279		7,256		3,187	

C Complementarity Willingness To Pay

Table C.6: Summary of savings, cost-minimizing plans, and predicted WTP across various regions and years.

Year	Region	Utility	Block rate savings	Cost-minimizing plan	Cost-minimizing plan savings	Model-predicted WTP for complementarity
2013	Sacramento	PG&E	2,688	EV rate	1,487	19,772
2013	San Francisco	PG&E	2,688	EV rate	673	13,587
2013	Central Valley	PG&E	2,688	EV rate	1,890	12,592
2013	Rest of State	PG&E	2,688	EV rate	1,352	19,803
2013	Los Angeles	SCE	2,272	EV rate	2,948	17,829
2013	San Diego	SDG&E	2,395	EV-TOU-2	429	15,555
2017	Sacramento	PG&E	1,316	EV rate	2,217	9,699
2017	San Francisco	PG&E	1,316	EV rate	2,644	7,306
2017	Central Valley	PG&E	1,316	EV rate	1,978	10,588
2017	Rest of State	PG&E	1,316	EV rate	2,397	9,643
2017	Los Angeles	SCE	1,248	EV rate	1,014	1,279
2017	San Diego	SDG&E	3,051	Block rate	3,051	6,306
2019	Sacramento	PG&E	878	EV rate	2,613	30
2019	San Francisco	PG&E	878	EV rate	2,378	4,202
2019	Central Valley	PG&E	878	EV rate	2,738	7,338
2019	Rest of State	PG&E	878	EV rate	2,523	11,581
2019	Los Angeles	SCE	783	EV rate	3,550	5,995
2019	San Diego	SDG&E	2,667	EV-TOU-5	1,577	10,915
Average Across Markets			1,775		2,081	10,223

Notes: The two savings columns are external tariff calculations rather than model output. The savings are discounted at three per cent over ten years; Appendix G gives the construction. The final column is model output: for each market, the amount by which an EV–PV bundle’s mean utility exceeds the sum of its components’, averaged over the two vehicle sizes, divided by each household’s own price coefficient and averaged across households using population weights. The last row is an unweighted average across the eighteen markets.

D Robustness Checks

The appendices in this section probe the assumptions the baseline model rests on: the exogeneity of charging-station placement, the absence of unobserved tastes common to both technologies, the choice of price instruments, and the pooling of survey waves.

1 Endogenous Charging-Station Placement

The baseline specification treats the county-level number of public EV charging stations as a demand shifter that enters the utility of EV alternatives. A potential concern is that charging infrastructure is not placed randomly: if unobserved preferences for EVs and solar panels are correlated, locations attractive to charging networks may also be locations with high unobserved solar demand, in which case the endogenous component of station placement could load onto the complementarity term Γ .

To address this concern, I use as an instrument the interaction of the pre-period (2012) density of local grocery stores and supermarkets (a proxy for the commercial locations where charging stations are typically sited) with the cumulative number of charging stations in all other California counties, a leave-out measure of the aggregate statewide rollout of charging infrastructure, following Li et al. (2017). The identifying assumption is that pre-determined grocery density shifts where stations get built as the statewide rollout expands, but does not directly affect household vehicle or solar demand conditional on controls.

Because station density is endogenous and enters a nonlinear discrete-choice utility function, I implement the instrument using the control-function approach of Petrin and Train (2010). I first regress station density on the instrument and exogenous controls and then include the resulting residual in utility.³⁷

Estimation I estimate the following first-stage equation for charging-station density:

$$S_{ct} = \pi_1 Z_{ct} + \mathbf{x}'_{ct} \boldsymbol{\pi}_2 + \lambda_t + \mu_r + v_{ct}, \quad (12)$$

where S_{ct} denotes the number of public charging stations per 1,000 residents in county c and year t . The instrument, Z_{ct} , interacts the county's 2012 grocery-store density with the cumulative number of charging stations in all other California counties in year t . The vector $\mathbf{x}_{ct}$ includes the county's share of population with a college degree, log median household income, and median age,

³⁷Li et al. (2017) use this variable as an excluded instrument for charging-station availability in their GMM estimation. In my nonlinear discrete-choice setting, I instead incorporate the instrument through a control function (Petrin and Train, 2010; Wooldridge, 2015), which uses the first-stage residual to control for the dependence between station density and unobserved preferences.

while λ_t and μ_r denote year and region fixed effects, respectively. Regions correspond to those used to define markets in the model.

The instrument strongly predicts charging-station density. The estimated coefficient on Z_{ct} is 0.196 ($t = 8.04$), with an excluded-instrument F -statistic of 64.7.

I then use the estimated residual from the first-stage regression as a control function for the endogeneity of charging-station density. The residual enters both EV and solar utility, allowing the endogenous component of station placement to be correlated with unobserved demand for either technology.³⁸ All other components of the estimation procedure, including the demographic interactions, contraction mapping, and second-stage IV estimation, are unchanged from the baseline specification.

³⁸This is important if charging infrastructure expands disproportionately in locations with stronger latent demand for green technologies, in which case controlling only EV utility would not fully address the resulting correlation between station placement and unobserved solar demand. I also test a specification where the control function enters EV utility only. The estimated complementarity remains essentially unchanged.

Table D.7: Robustness to Endogenous Charging-Station Placement

<i>First-stage parameters</i>		<i>Second-stage parameters</i>	
Income (50–100k) $\times$ prices	0.221 (0.023)	Constant	–0.163 (0.068)
Income (50–100k) $\times$ \$ per mile	–1.518 (0.637)	MPG_{EV}	–2.015 (0.193)
Income (100–200k) $\times$ prices	0.370 (0.023)	$MPG_{gas/hybrid}$	1.108 (0.281)
Income (100–200k) $\times$ \$ per mile	–3.004 (0.632)	Horsepower	2.989 (0.108)
Income (200k+) $\times$ prices	0.520 (0.024)	Car size	–0.410 (0.032)
Income (200k+) $\times$ \$ per mile	–5.468 (0.716)	PV	0.966 (0.645)
HH members $\times$ car size	–0.043 (0.003)	Complementarity, $\hat{\Gamma}$	1.365 (0.156)
College degree $\times$ mpg	2.216 (0.104)	Prices, $\hat{\alpha}$	–1.639 (0.058)
College degree $\times$ PV	0.041 (0.043)		
GHI $\times$ PV	1.930 (1.168)		
Age (35–64) $\times$ PV	–0.085 (0.067)		
Age (35–64) $\times$ EV	0.215 (0.130)		
Age (65+) $\times$ PV	0.289 (0.072)		
Age (65+) $\times$ EV	–0.024 (0.142)		
Station density $\times$ EV	0.174 (0.766)		
$\hat{v}_{ct} \times EV$	–0.011 (0.877)		
$\hat{v}_{ct} \times PV$	0.862 (0.381)		

Notes: The table reports estimates of the control-function specification. The control-function residual $\hat{v}_{ct}$ is estimated from equation (12); the excluded-instrument F -statistic is 64.7. Standard errors are reported in parentheses.

Results Table D.7 reports the results. The estimated complementarity parameter is essentially unchanged after accounting for endogenous charging-station placement: $\hat{\Gamma}$ declines from 1.366 in the baseline specification to 1.365, a difference within one standard error. The price coefficient and demographic interaction coefficients are similarly stable. The estimated control-function coefficient is positive for solar utility, consistent with the endogenous component of station placement being correlated with unobserved determinants of solar demand. Importantly, allowing for this correlation has virtually no effect on the estimated complementarity, suggesting that the baseline estimate of Γ is not driven by potentially endogenous charging-station placement.

2 Unobserved Correlated Preferences

The baseline model captures systematic correlation in household tastes for EVs and PVs through observable demographic interactions. To allow for common preferences not captured by these observables, I add a latent taste for “green” technologies to the individual-specific utility:

$$\tilde{\mu}_{imj} = \mathbf{X}'_{mj}\Phi\mathbf{D}_{im} + \sigma_g v_i G_j, \quad v_i \stackrel{iid}{\sim} N(0, 1), \quad (13)$$

where $G_j \in \{0, 1, 2\}$ is the number of green technologies contained in bundle j . The individual-specific random coefficient v_i generates correlated unobserved tastes for EVs and PVs. Thus, σ_g captures a common propensity to value both technologies, whereas Γ captures the additional utility from owning them jointly.

I estimate the extended first stage by simulated maximum likelihood using $S = 100$ draws per individual. The contraction mapping and second-stage IV estimation of Γ and α are otherwise unchanged.

Table D.8: Robustness to Correlated Unobserved Preferences

	Baseline	Correlated preferences
Complementarity, $\hat{\Gamma}$	1.366 (0.160)	1.343 (0.182)
Price coefficient, $\hat{\alpha}$	-1.641 (0.059)	-1.581 (0.058)
Latent green taste, $\hat{\sigma}_g$	—	0.209 (0.218)

Notes: Column 1 reports the baseline estimates from Table 1. Column 2 reports estimates from the specification in equation (13), estimated using simulated maximum likelihood with $S = 100$ draws per individual. Standard errors are in parentheses.

Table D.8 shows that allowing for a common unobserved taste for EVs and PVs leaves the

estimated complementarity largely unchanged. The estimate of Γ declines from 1.366 to 1.343, a shift of less than a sixth of a standard error, and remains statistically significant. The remaining coefficients are also quantitatively similar. The estimated standard deviation of the latent taste, $\hat{\sigma}_g = 0.209$, is imprecisely estimated, which is expected given that the demographic interactions already absorb systematic correlation in tastes. Therefore, the estimated complementarity is robust to allowing for a common unobserved taste for EVs and PVs.

3 Supply-Side Cost-Shifter Instruments

The baseline specification instruments for price using BLP-style sums of rival-product characteristics. A potential concern with these instruments is that rival-product characteristics may be correlated with unobserved demand shocks, particularly if unobserved preferences for green technologies are correlated across EV and solar panel choices. To assess the sensitivity of the estimates to this identifying assumption, I re-estimate the model replacing the baseline BLP instruments with alternative supply-side cost shifters. These instruments provide alternative sources of variation in the costs of supplying vehicles and installing solar.

Alternative instruments. I construct two sets of supply-side cost shifters, based on labor and input costs, and consider one policy instrument in addition. The labor-cost instruments consist of auto-dealer wages (NAICS 4411) interacted with an indicator for vehicle alternatives and construction wages (NAICS 23) interacted with the solar PV indicator, both taken from the Quarterly Census of Employment and Wages (U.S. Bureau of Labor Statistics, 2026b). Both wage measures vary at the county-year level. The input-cost instruments consist of battery-pack costs interacted with the EV indicator and the steel producer price index interacted with vehicle size. Battery-pack costs are annual lithium-ion pack prices from the BloombergNEF battery price survey (BloombergNEF, 2019), and the steel index is the producer price index for iron and steel, series WPU101 (U.S. Bureau of Labor Statistics, 2026a). These input-cost measures are observed at the national level and therefore vary over time but not across geographic markets within a given year. The policy instrument is PV rebates per watt interacted with the solar PV indicator, constructed from installation-level records in Lawrence Berkeley National Laboratory’s Tracking the Sun data (Lawrence Berkeley National Laboratory, 2024); it varies at the county-year level and enters only in the final specification. For the county-year measures (auto-dealer wages, construction wages, and PV rebates), I construct market values by taking population-weighted averages across counties within each market. This yields the corresponding measures for the 18 markets used in the estimation.

Table D.9 reports the results. Column (1) reproduces the baseline estimates using the BLP-style instruments. Column (2) replaces the baseline instruments with the labor-cost shifters, while column (3) uses the input-cost shifters. Column (4) combines the labor- and input-cost instruments,

and column (5) additionally includes the PV rebate instrument. Thus, columns (2)–(5) exclude the baseline BLP instruments entirely and identify the price coefficient using alternative sources of variation. The alternative instruments have substantial first-stage predictive power, with first-stage F -statistics ranging from 33.6 to 56.4. The estimated complementarity parameter is stable across these alternative instrument sets. The baseline estimate is $\hat{\Gamma} = 1.366$. Combining the labor- and input-cost instruments produces an estimate of $\hat{\Gamma} = 1.331$, and adding the PV rebate instrument gives $\hat{\Gamma} = 1.358$. Across all specifications that exclude the baseline BLP instruments, $\hat{\Gamma}$ therefore ranges from 1.250 to 1.389, compared with 1.366 in the baseline specification, and remains positive and statistically significant throughout. The estimated price coefficient also remains negative and statistically significant across all specifications. Overall, the results are robust to replacing the baseline BLP-style instruments with alternative labor- and input-cost shifters.

Table D.9: Robustness to Alternative Price Instruments

	(1) Baseline	(2) Wages	(3) Input costs	(4) Input+Wages	(5) Input+Wages + PV rebate
Complementarity, $\hat{\Gamma}$	1.366 (0.160)	1.250 (0.172)	1.389 (0.164)	1.331 (0.161)	1.358 (0.161)
Price coefficient, $\hat{\alpha}$	-1.641 (0.059)	-1.315 (0.121)	-1.704 (0.156)	-1.542 (0.094)	-1.617 (0.091)
First-stage F -statistic	393.7	39.6	33.6	56.4	48.7
Number of excluded instruments	5	2	2	4	5

Notes: This table reports estimates of the complementarity parameter Γ and the price coefficient α under alternative instrument sets for price. Column (1) reports the baseline specification, which instruments for price using BLP-style sums of rival-product characteristics. Columns (2)–(5) replace the baseline BLP instruments with alternative cost and policy shifters. Column (2) uses labor-cost shifters: auto-dealer wages (NAICS 4411) interacted with an indicator for vehicle alternatives and construction wages (NAICS 23) interacted with the solar PV indicator. Column (3) uses input-cost shifters: battery-pack costs interacted with the EV indicator and the steel producer price index interacted with vehicle size. Column (4) combines the labor- and input-cost instruments from columns (2) and (3). Column (5) additionally includes PV rebates per watt interacted with the solar PV indicator. Standard errors are reported in parentheses. The first-stage F -statistic tests the joint significance of the excluded instruments in the price equation.

4 Complementarity and Policy Conclusions by Adoption Cohort

Changes in technology and consumer preferences over 2013–2019 may affect the value of joint ownership. Appendix Table C.6 shows that willingness to pay for complementarity declines over the sample period. This pattern is consistent with stronger complementarity among earlier adopters, although comparisons across survey waves cannot distinguish changes in adopter composition from evolving preferences. I therefore compare the early-adopter cohort, surveyed in 2013, with the ones surveyed in 2017 and 2019, in order to assess whether these differences change the policy conclusions.

Table D.10 repeats the policy counterfactuals on each cohort’s markets alone, retaining the pooled preference coefficients and using the corresponding market-specific mean utilities, so that each exercise asks what the policy would achieve if all of California resembled that cohort’s respondents. The two cohorts differ in the strength of complementarity, but the policy conclusions are similar. Removing complementarity reduces joint ownership by 90% in the 2013 cohort and 60% in 2017–2019, and complementarity accounts for 34% and 28% of the carbon reductions attributable to EV subsidies. The absolute reductions run the other way: the same \$2 billion EV subsidy reduces CO_2 by 1.70 million metric tons in the later cohort against 1.03 in 2013, because the mean utility of both technologies is higher in the later markets, which puts more households near the margin of adoption and lets the same subsidy dollar induce more purchases. The benefit-maximizing allocation assigns 32% of the budget to PV in the 2013 cohort and 29% in 2017–2019. Thus, the early-adopter cohort exhibits stronger complementarity and the later cohort a larger policy payoff, while the optimal allocation differs between them by about three percentage points. Annual benefits change very little across allocations in this range.

Table D.10: Complementarity and Policy Conclusions by Adoption Cohort

	2013	2017–2019	Pooled
Bundle share without complementarity, % change	–90	–60	–64
EV-subsidy CO_2 reduction via complementarity, %	34	28	29
EV-subsidy $PM_{2.5}$ reduction via complementarity, %	53	42	43
Optimal PV share of budget, %	32	29	29

Notes: Each column reports counterfactuals computed on one group of markets alone: the 2013 cohort, the 2017–2019 cohorts, and the full sample. All three use the same pooled preference coefficients, combined with the mean utilities of the markets that column covers. The first row reports the percentage change in joint ownership when EV–PV bundle utility is set equal to the sum of its components. The next two rows report the percentage of EV-subsidy-induced reductions in CO_2 and $PM_{2.5}$ attributable to complementarity. The final row reports the share of the subsidy budget allocated to PV that maximizes monetized benefits.

E Two-Period Dynamic Extension

1 What the Static Estimator Recovers

I extend the two-good model in equations (7)–(10) to two periods to clarify how the static complementarity estimate should be interpreted when adoption is sequential and forward-looking. Households are forward-looking and discount Period 2 utility by $\beta \in (0, 1)$. Adoption is irreversible and Period 2 is terminal. A household pays the purchase price when it adopts a good; this price is sunk in the following period. Thus, a household may adopt both goods in Period 1 or adopt one good in Period 1 and add the other in Period 2.

Assume preferences and the flow utilities from ownership are constant across periods. In the baseline specification, Period 2 has the same choice set, product characteristics, and prices as Period 1. Idiosyncratic utility shocks are drawn independently in each period from the same Type I extreme-value distribution.

Define

$$z_A \equiv \delta_A + \alpha p_A \quad \text{and} \quad z_B \equiv \delta_B + \alpha p_B.$$

At the beginning of Period 2, a household can be in one of four ownership states. After integrating over the Period 2 idiosyncratic choice shocks, the corresponding continuation values are

$$V^{(AB)} = \delta_A + \delta_B + \Gamma, \tag{14}$$

$$V^{(A)} = \delta_A + \log \left(1 + e^{z_B + \Gamma} \right), \tag{15}$$

$$V^{(B)} = \delta_B + \log \left(1 + e^{z_A + \Gamma} \right), \tag{16}$$

$$V^{(0)} = \log \left(1 + e^{z_A} + e^{z_B} + e^{z_A + z_B + \Gamma} \right). \tag{17}$$

A household that owns one good chooses whether to add the other, whereas a household that owns neither retains the option to choose among all four alternatives. The Period 1 choice-specific values are therefore

$$v_0 = \beta V^{(0)}, \tag{18}$$

$$v_A = (1 + \beta)\delta_A + \alpha p_A + \beta \log \left(1 + e^{z_B + \Gamma} \right), \tag{19}$$

$$v_B = (1 + \beta)\delta_B + \alpha p_B + \beta \log \left(1 + e^{z_A + \Gamma} \right), \tag{20}$$

$$v_{AB} = (1 + \beta)(\delta_A + \delta_B) + \alpha(p_A + p_B) + (1 + \beta)\Gamma. \tag{21}$$

What does the static model recover? In the four-alternative static logit model, complementarity is recovered from the log-odds ratio

$$\widehat{\Gamma}^{\text{static}} = \log \left(\frac{P_{AB}P_0}{P_AP_B} \right), \quad (22)$$

where P_{AB} , P_A , P_B , and P_0 denote the static choice probabilities of the four alternatives. If these choice probabilities are generated by the dynamic model above, then

$$\begin{aligned} \widehat{\Gamma}^{\text{static}} &= v_{AB} + v_0 - v_A - v_B \\ &= (1 + \beta)\Gamma + C, \end{aligned} \quad (23)$$

where

$$C = \beta \log \left[\frac{1 + e^{z_A} + e^{z_B} + e^{z_A+z_B+\Gamma}}{(1 + e^{z_A+\Gamma})(1 + e^{z_B+\Gamma})} \right]. \quad (24)$$

The numerator minus the denominator inside the logarithm in equation (24) equals

$$(1 - e^{\Gamma}) \left(e^{z_A} + e^{z_B} + e^{z_A+z_B+\Gamma} \right).$$

Because the expression in parentheses is positive,

$$\text{sign}(C) = -\text{sign}(\Gamma).$$

Consequently, when $\Gamma > 0$,

$$\widehat{\Gamma}^{\text{static}} < (1 + \beta)\Gamma. \quad (25)$$

Thus, the static estimate is a lower bound on $(1 + \beta)\Gamma$, the two-period present discounted value of the true per-period complementarity flow Γ . The attenuation arises because delaying adoption preserves flexibility over future choices. A household that has not yet adopted either good retains the option to choose among all four alternatives in Period 2, whereas a household that has adopted one good can only retain it or add the other.

Accordingly, forward-looking behavior does not merely rescale the static complementarity parameter. The static estimate reflects the value of current and future joint ownership net of the option value associated with delaying adoption. It should therefore be interpreted as a measure of the value of joint ownership, rather than as a preference specifically for simultaneous adoption.

2 Policy Robustness to Forward-Looking Adoption

To understand the quantitative importance of forward-looking adoption for the policy counterfactuals, I re-estimate the full demand model using two-period dynamic choice probabilities for discount factors $\beta \in \{0.80, 0.90, 0.95\}$, and recover the implied one-period complementarity flow Γ .³⁹ The benchmark specifications assume stationary prices. I also re-estimate the model under three nonstationary scenarios in which households anticipate Period-2 prices that are 10% lower for EVs, PV systems, or both. These specifications use the benchmark discount factor $\beta = 0.95$.

Table E.11 reports the results. Under stationary prices, the re-estimated one-period complementarity flow ranges from 0.791 to 0.845. Although the per-period flow decreases with β , its two-period present discounted value, $(1 + \beta)\hat{\Gamma}$, increases from 1.521 to 1.542. These values are 11.3–12.9% larger than the static estimate of 1.366, consistent with the option-value attenuation characterized in equation (23).

The results are also insensitive to anticipated price declines. At $\beta = 0.95$, allowing Period-2 EV or PV prices to decline by 10% changes $(1 + \beta)\hat{\Gamma}$ from 1.542 to between 1.530 and 1.539, a difference of less than 1%. Across all specifications, the subsidy allocation that maximizes monetized emissions benefits assigns between 22% and 25% of the budget to PV subsidies, against 29% in the static model. Forward-looking adoption therefore shifts the optimum modestly toward EVs, by roughly four to seven percentage points, and by less the more heavily the future is discounted. The static allocation remains within 0.4% of the best attainable benefit in every dynamic specification, so the policy conclusion is unaffected.⁴⁰ Thus, re-estimating demand to allow for forward-looking adoption changes the magnitude of the complementarity associated with joint ownership while leaving the optimal subsidy allocation broadly stable.

³⁹I use $\beta = 0.95$ as the benchmark, consistent with the standard annual discount factor in dynamic models of durable-good demand (Bayer et al., 2016). The alternatives $\beta = 0.90$ and $\beta = 0.80$ capture either longer effective period lengths (approximately two and four years under annual discounting of 0.95) or the higher implicit discount rates estimated for residential energy investments. For example, De Groote and Verboven (2019) estimate an annual discount rate of approximately 15% among solar adopters. The results vary only modestly across these values.

⁴⁰The policy counterfactual is recomputed using each specification's complete re-estimated parameter vector.

Table E.11: Policy Robustness to Forward-Looking Adoption

Specification	β	Per-period flow $\hat{\Gamma}$	Comparable complementarity	Optimal PV subsidy share
Static baseline	–	–	1.366	29%
Stationary prices	0.80	0.845 (0.099)	1.521	25%
Stationary prices	0.90	0.808 (0.095)	1.536	22%
Stationary prices	0.95	0.791 (0.093)	1.542	22%
10% lower Period-2 EV price	0.95	0.789 (0.093)	1.539	22%
10% lower Period-2 PV price	0.95	0.786 (0.093)	1.532	22%
10% lower Period-2 EV and PV prices	0.95	0.785 (0.092)	1.530	22%

Notes: For the static baseline, the comparable complementarity measure is $\hat{\Gamma}^{\text{static}}$. For the dynamic specifications, it is the two-period present discounted value $(1 + \beta)\hat{\Gamma}$. Standard errors are reported in parentheses. The final column reports the share of the fixed subsidy budget allocated to PVs by the allocation that maximizes annual monetized benefits from reductions in CO₂ and PM_{2.5}, recomputed under each specification's re-estimated parameters.

F Complementarity Mechanism

Many utilities offer plans with a baseline allowance for monthly electricity usage. The baseline allowance is the maximum monthly energy use allowed at the lowest tier rate. Customers will advance to the higher tier price and beyond if they consume more than the allotted amount throughout the billing cycle. Customers who are on a Standard or Time-of-Use plan are affected by this. However, many utility companies offer electricity plans specifically for households with electric cars, and these plans usually do not have a baseline allowance.

In Table F.12, I provide two different electricity plans' rates: for EV customers only (EV-TOU-2) and for all customers (TOU-DR1). If we compare rates, we can see that TOU-DR1 has lower prices per kWh during all hours if the household uses below 130% of the baseline allowance compared to the rates in the EV plan. Therefore, installing a solar panel will help deal with the increased energy consumption from the electric car and stay at tier 1, which provides the lowest rates possible. I give the example of three possible scenarios of household electricity usage in Figure F.3. In the

first instance, before and after purchasing an electric vehicle, the household consumes less than 130% of the baseline allotment. Therefore, such households lack the motivation to purchase solar to maintain their lower rate tier. Meanwhile, types 2 and 3 will benefit from installing solar panels to stay below 130% of the baseline. For type 2 households, the objective is to cut solely the extra power used for EV charging, whereas, for type 3 households, the goal is to reduce both the extra electricity used for EV charging and the initial excess demand in consumption.

Another way to get the additional savings from complementarity is to take advantage of EV-specific time-of-use (TOU) rate structures. These EV TOU rates, available only to households with an EV, enable households to sell excess solar energy back to the grid at higher rates during peak solar generation hours. They can then use energy at a lower cost during off-peak hours, such as at night when they charge their EVs. This setup allows customers with both EVs and PV systems to generate extra income by optimizing their energy generation and consumption patterns.

Table F.12: Example of Electricity Rates in SDG&E (Summer 2022)

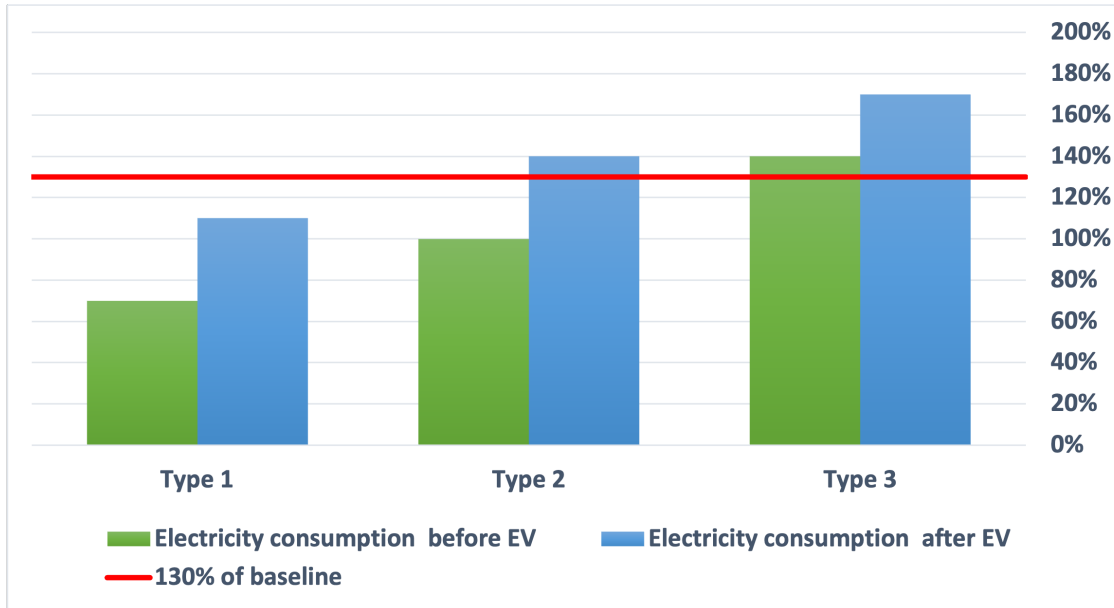
On-Peak	Super Off-Peak	Off-Peak
4:00 pm - 9:00 p.m.	Midnight –6 : 00 a.m. Midnight - 2:00 p.m. (Weekends & Holidays)	All other hours
EV-TOU-2		
66.7¢	23.8¢	41.9¢
TOU-DR1		
Tier 1(Up to 130% baseline)		
58.5¢	23.4¢	35.6¢
Tier 2(> 130% baseline)		
69¢	33.6¢	45.8¢

Source: San Diego Gas & Electric Company (2022)

G Supporting Details for Complementarity Savings Calculations

To compute savings from the complementarity channel in different regions of California, I use several data sources. I begin with the 2014 average daily load shape for California residential customers reported by California Energy Commission (2019). Additionally, data from the 2020 Residential Energy Consumption Survey (RECS) by U.S. Energy Information Administration (2020) indicate that single-family homes with solar panels and electric vehicles in California consume an average

Figure F.3: Electricity consumption for different types of households



of 8,972 kWh annually, or 24.58 kWh per day. Of this, 1,749 kWh per year (4.79 kWh/day) is attributed to EV charging.

To estimate hourly electricity consumption, I subtract EV charging demand from the total daily consumption and distribute the remaining load according to the average daily load shape. EV charging is assumed to occur during off-peak hours (11:00 PM to 6:00 AM) in line with typical EV-TOU rate structures that offer lower prices during these hours, supported by Burlig et al. (2021).

For average daily solar generation, I calculate the median system size (in kW_{DC}) for each region and year using data from Lawrence Berkeley National Laboratory (2023). I adjust for regional differences in global horizontal irradiation rates using Solargis (2022) and apply system performance adjustments based on derating factors and inverter efficiencies as documented by Franklin (2019). Solar generation is assumed to occur between 10:00 AM and 5:00 PM, consistent with net load reduction periods identified by California Independent System Operator (2016).

Finally, I estimate the approximate electric bill for the average customer with these characteristics under various regional rate plans.

H Assumptions Used for Calculations

- Adult population (18 years and older) of California is 30,827,105 (U.S. Census Bureau, 2021). About 67% of this population live in one unit houses or mobile homes according to Flood et al. (2023). Therefore, the potential market size is about 20,654,160 people.
- According to Lawrence Berkeley National Laboratory (2024), the median solar system size

installed in California is 6.6 kW, which will generate approximately 9,000 kWh of electricity per year.

- California's electricity generation CO_2 equivalent emissions are 457.5 lb/MWh (U.S. Environmental Protection Agency, 2024b).
- The average annual emissions for a gasoline vehicle in California, including both upstream and tailpipe emissions, are 12,594 lbs of CO_2e . A hybrid car's annual emissions are 6,898 lbs of CO_2e . For battery electric vehicles, annual emissions are 1,385 lbs of CO_2e , while for plug-in hybrid electric vehicles (PHEVs), total emissions are 3,866 lbs of CO_2e (U.S. Department of Energy, 2024). Since in my estimation, I treat both PHEV and BEV as electric vehicles, I will assume that average EV emissions charged from the grid are 2,626 lbs (1,191 kg) per year.
- An EV owned jointly with a PV system is assumed to charge from the grid and is assigned the same emissions as an EV owned alone.
- $PM_{2.5}$ emissions from electricity generation in California are 0.0281 lb/MWh (U.S. Environmental Protection Agency, 2023). Thus, a PV system reduces $PM_{2.5}$ emissions by $9,000 \times 0.0281 / 1,000 = 0.2529$ lbs (0.115 kg) per year.
- A PV system therefore displaces $9,000 \times 457.5 / 1,000 = 4,118$ lbs (1,868 kg) of CO_2 per year. The model's choices are made by individual respondents, whereas a PV system serves the whole household and every member of a household reports the same PV status, so a respondent-level share counts one system once per member. The household-level factors are therefore divided by the number of sampled persons per household, 2.18 (19,859 respondents in 9,096 households). The factors applied per respondent are 856 kg of CO_2 and 0.053 kg of $PM_{2.5}$.
- The average annual vehicle miles of travel (VMT) is 14,489 (U.S. Department of Transportation, 2022).
- $PM_{2.5}$ emissions from cars are 0.003 grams per mile or 0.043 kg per year given the average mileage (U.S. Environmental Protection Agency, 2024a).

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